ATRIA

HEALTHCARE APPOINTMENTS

TECHNICAL DOCUMENT

How Atria is built.

The architecture, data model, data flows, behaviour and authorisation of the Atria healthcare appointment booking and reminder system, with every UML, data flow and architecture diagram drawn for it and the decisions each one encodes.

DOCUMENT	REVISION	DATE	STATUS
ATR-TD-001	Rev B	22 September 2026	Issued for review

PROJECT	TEAM	STANDARD	DIAGRAMS
DIGISOL Internship 2026	Bryan · Thanksgiving	ISO/IEC/IEEE 42010	20 figures

DOCUMENT CONTROL

REV	DATE	CHANGE	AUTHOR
A	17 September 2026	First issue. Architecture, data flow diagrams Level 0 to 2, UML use case, class, component, sequence and state machine diagrams, entity relationship diagram, authorisation model, interface contract and operations.	Bryan
B	22 September 2026	Revised during the first build. Adds the platform stack that publishes the shared Lambda dependency layer, records that the pre-token generation trigger is the V2 form because only that version can add claims to an access token, states the Lambda packaging and continuous integration gates, and carries the redrawn state machine diagrams UML-07a to 07c with orthogonal transitions.	Bryan

PREPARED	REVIEWED	APPROVED
Project team	Internship supervisor	DIGISOL

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1 Introduction

1.1 Purpose

This document describes how Atria is built: the parts of the system, how data moves between them, what is stored and who may read it, and how the system behaves over time. It is written for the engineers who build and operate Atria and for the reviewers who assess it. Every diagram of the system is reproduced here in full, with the explanation needed to read it.

1.2 The system in one paragraph

Atria lets patients in Cameroon book, follow and cancel outpatient visits, reminds them by SMS, and gives clinic staff the tools to run the day, including for patients who never install anything. It runs as a serverless AWS backend behind one REST API, with a Flutter app for patients and staff and a React web app for every role. Tenants are institutions such as a clinic group or a health district; every record belongs to exactly one tenant.

1.3 Scope

COVERED	NOT COVERED
Architecture, components and infrastructure as code	Visual design, screens and interaction
Data flows at Levels 0, 1 and 2	Clinical records, prescriptions and results
Data model, keys and retention	Payments and settlement (the ledger records entitlement only)
Behaviour: sequences and state machines	Integration with the national health information platform
Authorisation model and interface contract	Tenant onboarding commercial terms

1.4 Notation

DIAGRAM	NOTATION USED
Data flow diagrams	Gane and Sarson: rounded rectangles with a number band for processes, open-ended rectangles with an ID box for data stores, plain rectangles for external entities, labelled arrows for data flows. A store or entity drawn twice carries a duplicate mark.
Use case	UML 2.5 actors, use cases, system boundary, include and extend dependencies.
Class	UML 2.5 classes with attributes and operations, generalisation, composition, aggregation and multiplicities.
Component	UML 2.5 components with provided (ball) and required (socket) interfaces and dependencies.
Sequence	UML 2.5 lifelines, synchronous messages as solid arrows, asynchronous messages as open arrows, returns as dashed arrows.
State machine	UML 2.5 initial and final pseudostates, simple and composite states, labelled transitions.

DIAGRAM	NOTATION USED
Entity relationship	Crow's foot cardinality: exactly one, zero or one, zero or more.
Architecture	AWS Architecture Icons 2024, grouped by layer.

1.5 Terms

TERM	MEANING
Tenant	An institution using Atria. Every record belongs to exactly one tenant.
Service line	General consultation, specialist consultation or procedure. Each is booked differently.
Session	A published window of general consultation capacity at a clinic, with a number of rostered GPs.
Queue ticket	A patient's place in a session, with a server-assigned position and a live arrival window.
Grid unit	The calendar resolution set by the tenant, 10 minutes by default.
Slot lock	The record whose existence reserves one grid unit of a clinician's time.
Care context	The intake answers a patient sends ahead of one visit, stored apart from the appointment.
Region pack	Country data: currency, phone rules, languages, timezone, holidays, formats and fee defaults.
Entitlement	The refund band an appointment outcome gives the patient: NONE, PARTIAL or FULL.
Person	One human being, holding no role. Patient profiles and staff memberships hang off a person.

1.6 Diagram register

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SHEET	DIAGRAM	SECTION
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2 Context, constraints and decisions

Four constraints of the Cameroonian setting shape the architecture. Each is met by one or more of the design decisions that follow.

ID	CONSTRAINT	WHAT IT REQUIRES	MET BY
C-01	Staffing	Most facilities do not have two doctors on site, and emergency cover is the scarcest resource. The platform must never promise a named doctor at a minute for general consultations.	AD-02, AD-03
C-02	Professional conduct and fees	Fees follow a ministerial schedule and publicity about named doctors is prohibited. Ratings must not change prices or appear on a named clinician's listing.	AD-11
C-03	Personal data law	Law No. 2024/017 governs health data, requires prior authorisation for processing and for transfer abroad, and places minors' sensitive data under prior authorisation. Development runs on synthetic data only.	AD-06, AD-09, AD-13, section 5.4
C-04	Connectivity	SMS is the primary reminder channel because mobile subscriptions far exceed internet use. Clients degrade gracefully offline.	AD-02, section 5.6

2.1 The care model

LINE	BOOKED AGAINST	WHO BOOKS	TIME SHOWN TO THE PATIENT
General consultation	A place in a session	Patient or staff	An arrival window and a queue position
Specialist consultation	A named clinician and a real slot	Patient or staff	A start time and the type's duration
Procedure	A named clinician or room, sized by staff	Staff; patients may request	None until staff place it and the patient confirms

Why the general line has no doctor and no minute.
At most facilities one doctor covers both walk-in and emergency care, so a promise of a named doctor at a set time is the promise most likely to break. A session queue keeps the clinic's working day unchanged and stops wasted trips, which is the problem reminders exist to solve.

2.2 Design decisions

ID	DECISION	WHAT IT MEANS FOR THE BUILD
AD-01	Three service lines	General consultation, specialist consultation and procedure are booked differently. Check-up and follow-up are appointment types inside a line, not lines of their own.
AD-02	Book a place in a session	The general line books a queue ticket in a published session. The patient gets a position and an arrival window, and the doctor is assigned at check-in by whoever is free.

ID	DECISION	WHAT IT MEANS FOR THE BUILD
AD-03	Disruption offers, it does not impose	When a clinician is marked unavailable, affected specialist appointments move to DISRUPTED and the patient is offered the next matching slots with one tap to confirm. Opt-in auto-accept applies to same-day slots only.
AD-04	Grid unit plus typed duration	The tenant sets a calendar grid unit (default 10 minutes) and every appointment type carries its own duration and buffer in grid units. A booking locks N consecutive units in one transaction.
AD-05	Procedures are staff-sized	A patient can request a procedure, which creates a REQUESTED appointment with no time held. Staff size and place it, and the patient confirms.
AD-06	Intake per appointment type	Each type carries its own intake form. Answers are stored as care context: a separate entity, encrypted under its own key, readable only by the patient and the assigned clinician, and purged sooner than booking history.
AD-07	Fee ledger without moving money	Every appointment writes a ledger entry at closure with fee band, amount in XAF, outcome and refund entitlement. Nothing settles until a payment provider is added. No-show strikes are tenant-configurable and off by default.
AD-08	Four range views and a history horizon	Day, week, month (density grid) and year (twelve-month heatmap). Records are retained ten years and shown to patients five years by default, subject to confirmation (OI-12).
AD-09	Date of birth and a verified phone	Every patient confirms a Complete your profile step whatever the identity provider returns. Date of birth is stored, age is derived. The phone is verified by one-time code. Under 18 requires a linked guardian with recorded consent.
AD-10	One person, separate profiles, fair queues	A person holds a patient profile and, separately, staff memberships. Staff scope is dropped when the actor is the subject. Queue order is assigned by the server. Self-benefiting reschedules are blocked, every staff reschedule is reasoned and audited, and staff priority exists only as a granted quota.
AD-11	Ratings stay off prices and off names	Feedback is collected after attended visits and seen by the institution, never shown on a patient-facing clinician listing. Fees have a per-practice baseline. The ratingAffectsFee flag exists and is refused by the Cameroon region pack.
AD-12	Channel and type are two axes	Every booking records how it reached the clinic (online, walk-in, phone, referral) separately from what is booked. Referral opens referring facility and clinician fields and restricts the types to specialist and procedure.
AD-13	Region pack	Currency, phone rules, languages, timezone, holidays, address and ID formats, SMS sender policy, emergency number and default fees come from a region pack chosen at onboarding and locked after the first appointment. Cameroon: XAF, +237, French and English, Africa/Douala.
AD-14	Queues, dead letters, alarms and a budget	Every asynchronous consumer reads from SQS with a dead letter queue. CloudWatch alarms, X-Ray on the booking path, AWS Budgets with anomaly detection, CloudTrail and customer-managed KMS keys are part of the baseline in every environment; WAF is added in production only.
AD-15	Roles are fixed at account creation	A tenant administrator creates each staff account with its roles. A staff member signs in to that account and receives exactly the permissions of its roles. There is no role switching in a session; a person who is also a patient uses the patient app with a separate sign-in.

3 Use cases

The use case diagram shows what each actor does with Atria. Staff actors are separate roles, each created by a tenant administrator and signed in to an account that carries only that role's permissions.

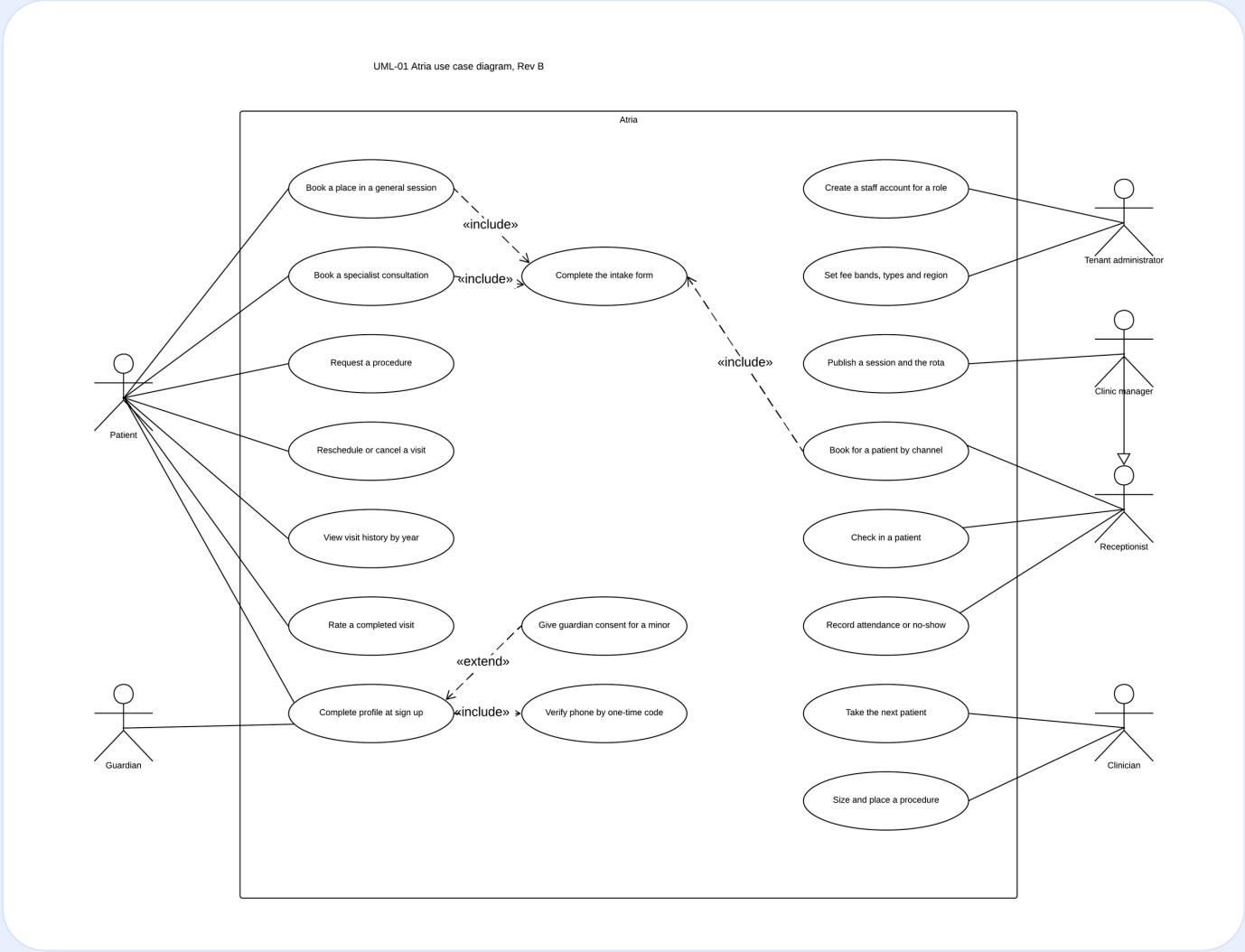


Figure 1. UML-01 use case diagram. Patient and guardian on the left; receptionist, clinician, clinic manager and tenant administrator on the right; include dependencies for the intake form and guardian consent.

ACTOR	ROLE	LANDS ON AFTER SIGN-IN	NOTES
Patient	PATIENT	Upcoming visits	Held through a patient profile, never through a staff membership. A staff member acting on their own appointment gets this role only.
Receptionist	RECEPTIONIST	Today at this clinic	The front desk. Runs the day, never sees clinical context.
Clinician	CLINICIAN	My calendar	A bookable health professional. Sees care context for their own patients and nobody else's.
Clinic manager	CLINIC_MANAGER	Clinic overview and rota	Runs one clinic. Adds the rota, reporting and the reschedule audit list on top of the receptionist's day.

ACTOR	ROLE	LANDS ON AFTER SIGN-IN	NOTES
Tenant administrator	TENANT_ADMIN	Tenant overview	Runs the institution. Deliberately cannot read care context.

SECONDARY ACTOR	PART IN THE SYSTEM
Guardian	Consents for a patient under 18 and receives reminders by default.
Google identity provider	Optional patient sign-in; birthday and phone requested as optional extra permissions.
SMS carrier	Delivers reminders, confirmations and queue-moved notices on MTN, Orange and Camtel.
Email provider	Delivers confirmation and cancellation email and returns delivery events.
Scheduler	Fires reminder and feedback-prompt events at their due time.

4 Data flow diagrams

The data flow diagrams show what data enters, leaves and is stored by Atria, independent of how it is built. Level 0 treats the system as one process. Level 1 opens it into seven processes and ten stores. Level 2 opens the four processes with the most internal structure. Each lower level balances with its parent: every flow that crosses a process boundary appears at both levels, or splits into named parts.

4.1 Notation and rules

Process
Rounded rectangle. The number band identifies it across levels.

Data store
Open-ended rectangle. The ID box carries D1 to D10.

External entity
Plain rectangle. A source or destination outside the system.

Data flow
Labelled arrow. The label names the data, not the action.

- Data moves only between a process and something else. No flow connects an entity to a store or a store to a store directly.
- No two flows cross, and no flow passes through a shape it does not connect.
- Where a flow at one level is a composite, the level below names its parts, and the diagram's note says so.
- A store or entity drawn twice to avoid crossings carries a duplicate mark: an extra bar in a store's ID box, a diagonal line in an entity's corner.

4.2 Level 0, context

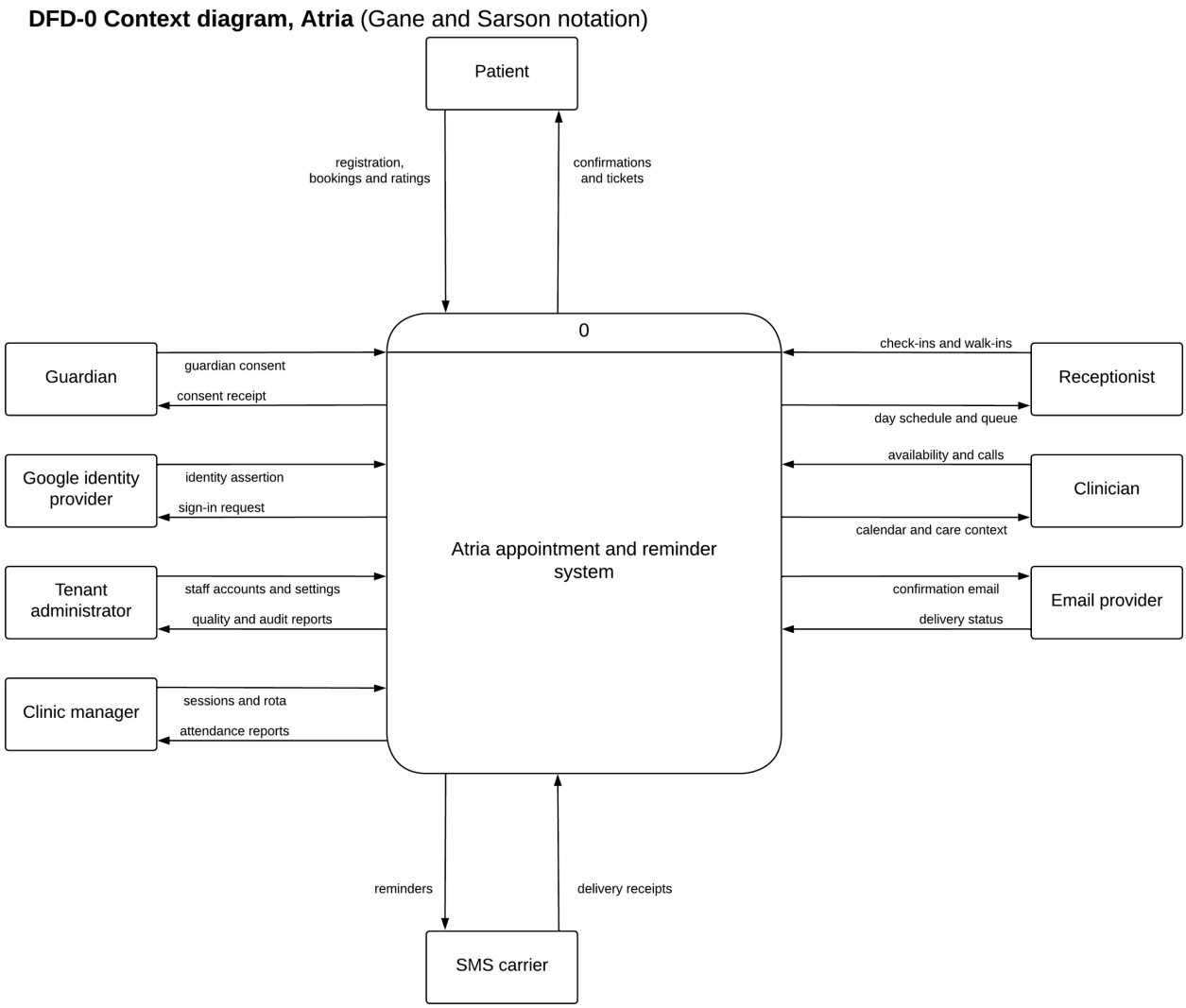


Figure 2. DFD-0 context diagram. One process and nine external entities.

ENTITY	DIRECTION	FLOW	CONTENT
Patient	into Atria	registration, bookings and ratings	Account details, booking and procedure requests, intake answers, post-visit ratings
Patient	from Atria	confirmations and tickets	Booking confirmations, queue tickets with position and arrival window, disruption offers
Guardian	into Atria	guardian consent	Consent for a patient under 18, with who, when and method
Guardian	from Atria	consent receipt	Confirmation of the consent recorded
Google identity provider	from Atria	sign-in request	OAuth authorisation request for a patient choosing Google

ENTITY	DIRECTION	FLOW	CONTENT
Google identity provider	into Atria	identity assertion	Signed identity token with name and email
Tenant administrator	into Atria	staff accounts and settings	Staff accounts with roles, clinics, appointment types, fee bands, region pack, feature switches
Tenant administrator	from Atria	quality and audit reports	Feedback analytics, audit trail, tenant reports
Clinic manager	into Atria	sessions and rota	Published sessions with rostered GPs
Clinic manager	from Atria	attendance reports	Attendance, no-show by channel, weekly reschedule list
Receptionist	into Atria	check-ins and walk-ins	Arrivals, walk-in bookings, phone bookings
Receptionist	from Atria	day schedule and queue	The clinic day and the merged live queue
Clinician	into Atria	availability and calls	Working hours, unavailability windows, next-patient calls
Clinician	from Atria	calendar and care context	Own calendar and the care context of own patients
SMS carrier	from Atria	reminders	Reminder, confirmation and queue-moved SMS
SMS carrier	into Atria	delivery receipts	Carrier delivery status per SMS
Email provider	from Atria	confirmation email	Confirmation and cancellation email
Email provider	into Atria	delivery status	Delivery, bounce and complaint events

4.3 Level 1, system

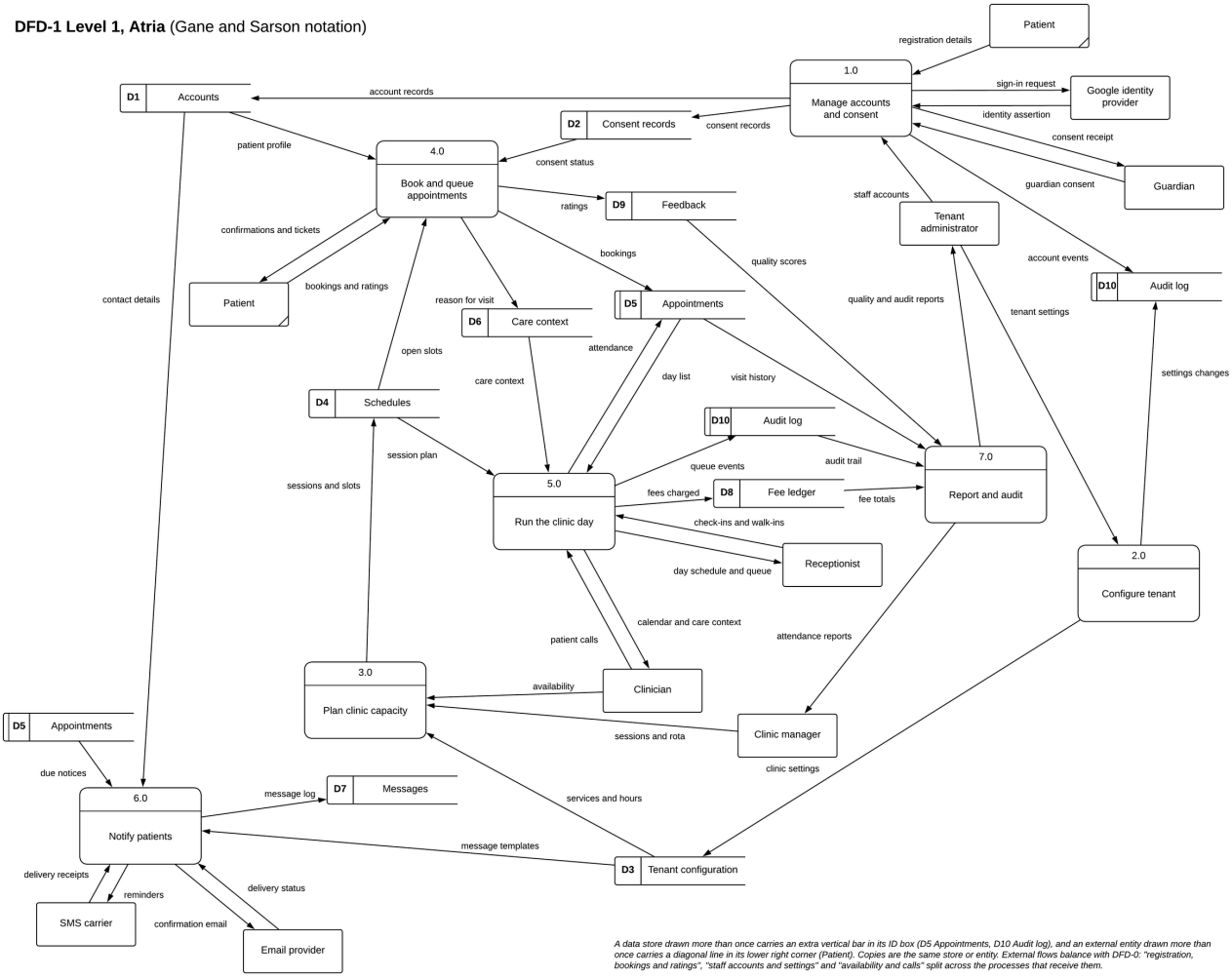


Figure 3. DFD-1 system diagram. Seven processes, ten data stores and the Level 0 entities. D5 Appointments, D10 Audit log and the Patient entity are drawn twice, with duplicate marks.

PROCESS	NAME	RESPONSIBILITY
1.0	Manage accounts and consent	Registers patients and signs them in, provisions staff accounts with their roles, and records patient and guardian consent.
2.0	Configure tenant	Maintains clinics, appointment types, intake forms, fee bands, the region pack and feature switches.
3.0	Plan clinic capacity	Turns availability, sessions, rota and services into bookable slots and published sessions.
4.0	Book and queue appointments	Validates requests, assigns slots or queue tickets, stores intake answers, confirms to the patient and records ratings.
5.0	Run the clinic day	Plans the day, receives and queues patients, runs consultations and closes visits with their fees.
6.0	Notify patients	Composes reminders and confirmations, sends SMS and email, and tracks delivery.

PROCESS	NAME	RESPONSIBILITY
7.0	Report and audit	Produces attendance, quality and audit reports from visits, fees, feedback and the audit log.

STORE	NAME	HOLDS (ENTITIES IN SECTION 6)
D1	Accounts	Person, patient profile, staff membership, clinician profile
D2	Consent records	Consent record
D3	Tenant configuration	Region pack, tenant, clinic, appointment type, intake form, fee band, priority allowance
D4	Schedules	Availability template, availability exception, session, slot lock
D5	Appointments	Appointment, queue ticket, appointment event, referral
D6	Care context	Care context, on its own key
D7	Messages	Reminder, message log
D8	Fee ledger	Fee ledger entry
D9	Feedback	Feedback
D10	Audit log	Audit entry

Balancing with Level 0.
Three Level 0 flows are composites and split across the processes that receive them: registration, bookings and ratings becomes registration details (1.0) and bookings and ratings (4.0); staff accounts and settings becomes staff accounts (1.0) and tenant settings (2.0); availability and calls becomes availability (3.0) and patient calls (5.0). Every other flow carries the same label at both levels.

4.4 Level 2

PROCESS 1.0, MANAGE ACCOUNTS AND CONSENT

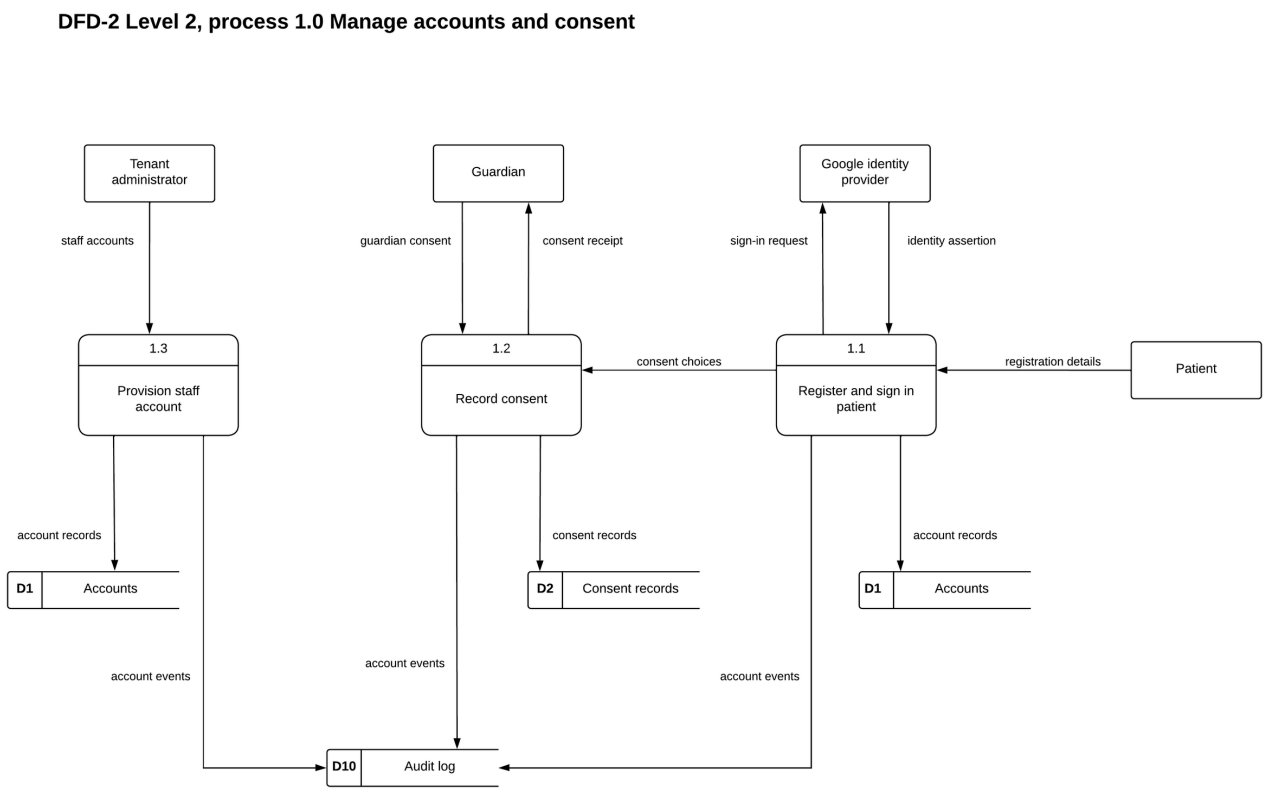


Figure 4. DFD-2 for process 1.0, manage accounts and consent.

PROCESS	NAME	RESPONSIBILITY
1.1	Register and sign in patient	Takes registration details, runs Google sign-in where chosen, writes the account and passes consent choices on.
1.2	Record consent	Records the patient's consent and, for a minor, the guardian's consent, and returns a receipt.
1.3	Provision staff account	Creates a staff account with the roles the tenant administrator assigns.

Every sub-process writes account events to the audit log. The Level 1 flow account records is written by both 1.1 and 1.3.

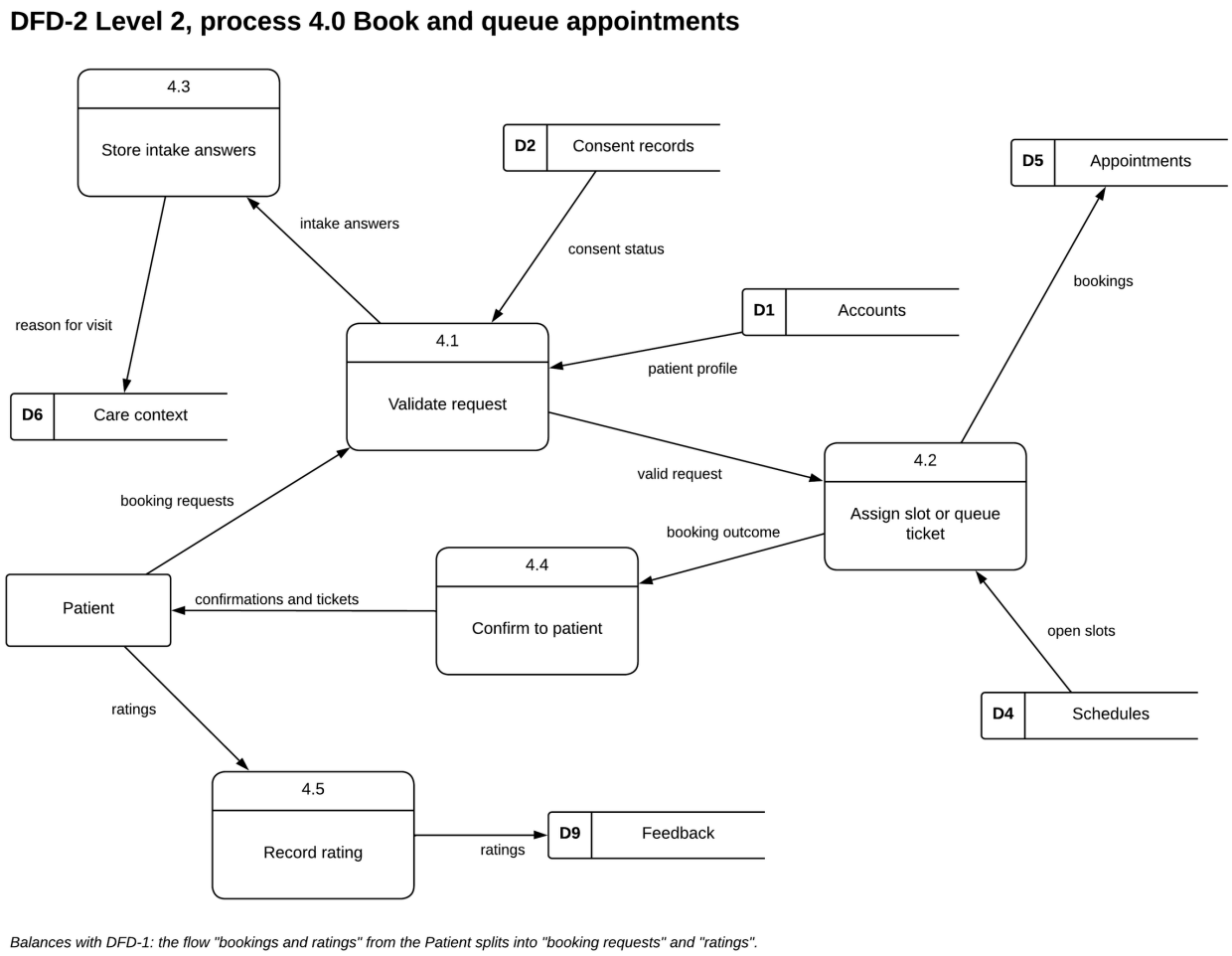


Figure 5. DFD-2 for process 4.0, book and queue appointments.

PROCESS	NAME	RESPONSIBILITY
4.1	Validate request	Checks the patient profile and consent status and splits the request into the booking and its intake answers.
4.2	Assign slot or queue ticket	Locks the slot units for a specialist booking or issues a server-ordered ticket in a session.
4.3	Store intake answers	Writes the intake answers to care context under the care context key.
4.4	Confirm to patient	Returns the confirmation or the ticket with position and arrival window.
4.5	Record rating	Stores a post-visit rating in feedback.

The Level 1 flow bookings and ratings splits here into booking requests and ratings.

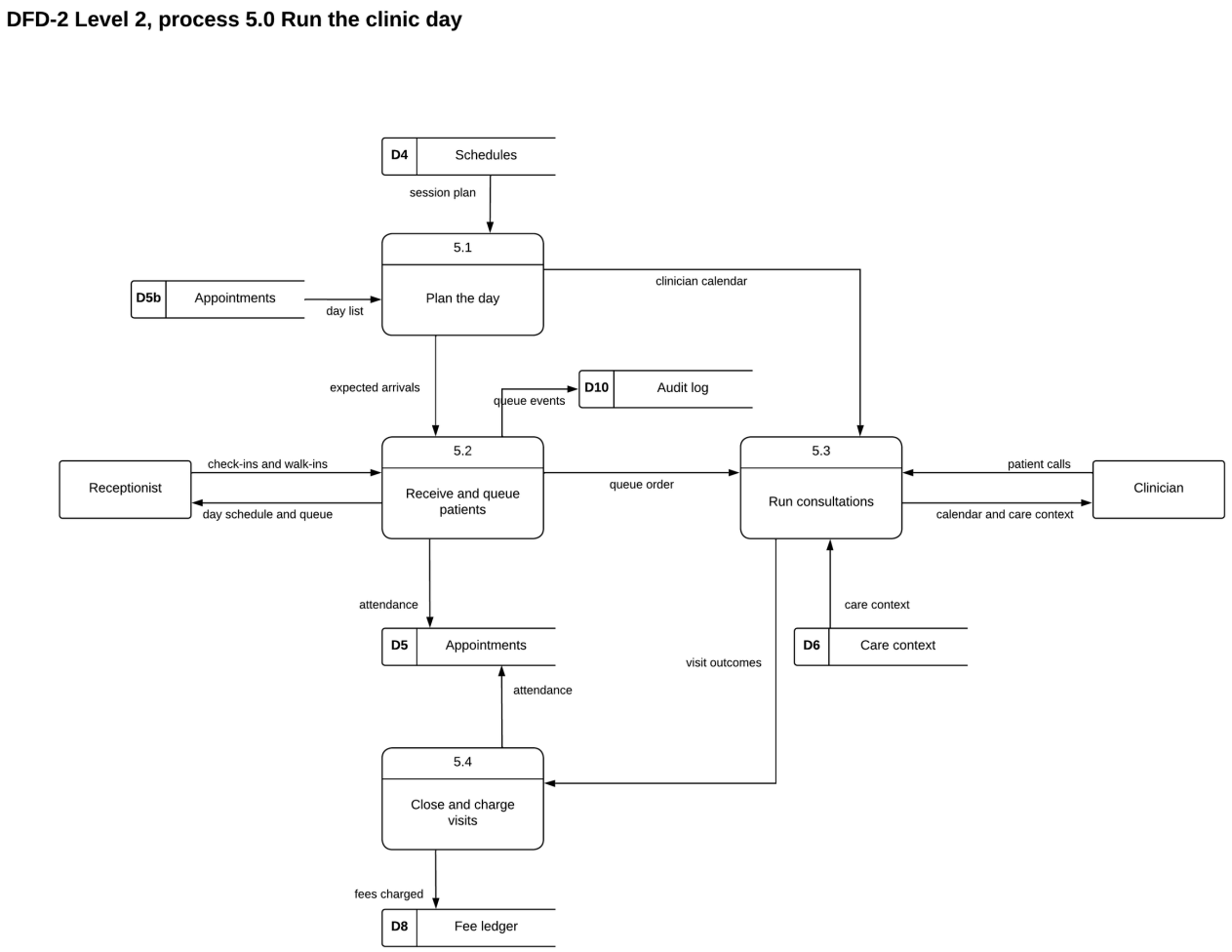


Figure 6. DFD-2 for process 5.0, run the clinic day.

PROCESS	NAME	RESPONSIBILITY
5.1	Plan the day	Combines the session plan and the day list into expected arrivals and each clinician's calendar.
5.2	Receive and queue patients	Records check-ins and walk-ins, orders the merged queue and writes attendance and queue events.
5.3	Run consultations	Serves the next patient to the calling clinician with the calendar and care context, and records visit outcomes.
5.4	Close and charge visits	Records the final attendance outcome and writes the fee ledger entry.

Attendance is written by 5.2 on arrival and by 5.4 at closure.

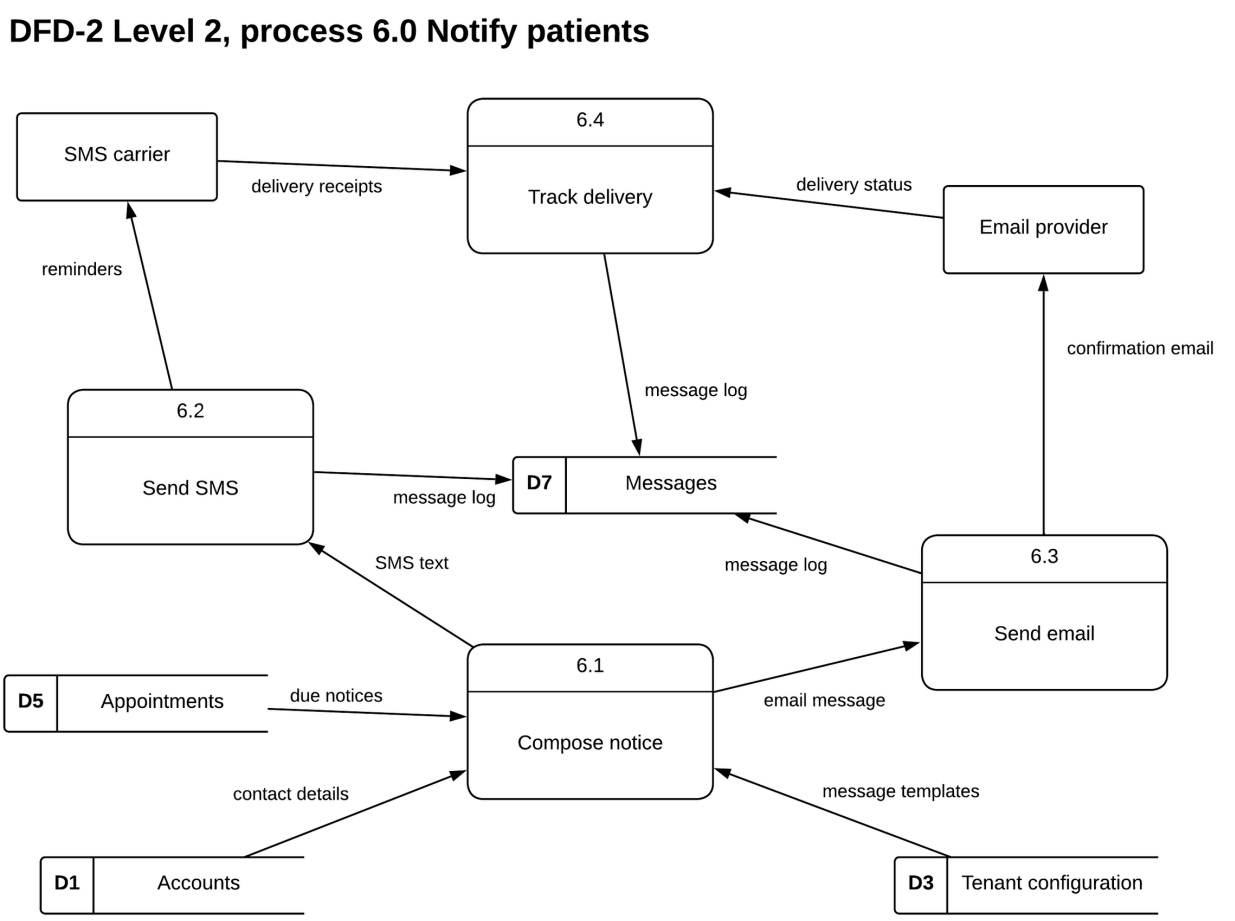


Figure 7. DFD-2 for process 6.0, notify patients.

PROCESS	NAME	RESPONSIBILITY
6.1	Compose notice	Builds each notice from the due appointment, the message template and the contact details.
6.2	Send SMS	Sends reminders through the SMS carrier and logs the message.
6.3	Send email	Sends confirmation email and logs the message.
6.4	Track delivery	Applies delivery receipts and delivery status to the message log.

Each sending and tracking sub-process writes the message log.

5 Architecture

5.1 Deployment architecture

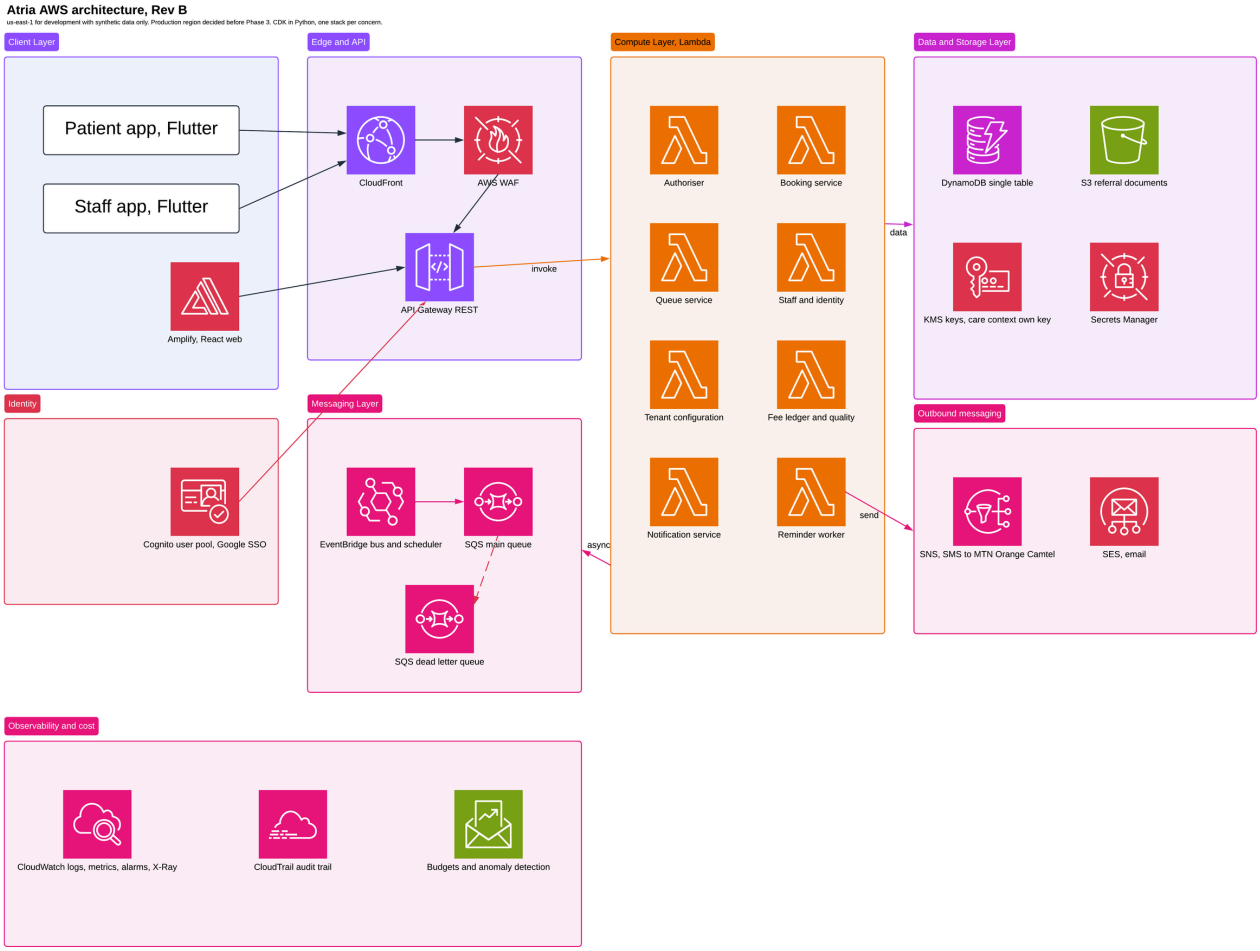


Figure 8. ARCH-2 AWS architecture. Clients reach CloudFront and WAF in front of API Gateway; Lambda services read and write DynamoDB, S3 and KMS; EventBridge and SQS carry asynchronous work to the reminder worker, which sends through SNS and SES. Observability and cost controls span every environment.

LAYER	SERVICES	ROLE
Clients	Flutter patient app, Flutter staff app, React web on Amplify Hosting	Patient and staff on phones and tablets; every role on the web; administration on desktop web only
Edge and API	CloudFront, AWS WAF, API Gateway (REST)	TLS termination, rate limiting and managed rule sets, request validation, one API for all clients
Identity	Cognito user pool, Google as identity provider	Accounts, sign-in, one-time codes, token issue with tenant and role claims
Compute	Lambda: authoriser and service functions	Stateless handlers, one IAM role each, least privilege

LAYER	SERVICES	ROLE
Data and storage	DynamoDB, S3, KMS, Secrets Manager	Single-table main store, care context table on its own key, referral documents, third-party secrets
Messaging	EventBridge bus and Scheduler, SQS with dead letter queues	Reminder schedules, event fan-out, FIFO outbox per appointment
Outbound	SNS (SMS), SES (email)	Reminders and confirmations with delivery events
Observability and cost	CloudWatch, X-Ray, CloudTrail, Budgets, Cost Anomaly Detection	Logs, metrics, alarms, traces, audit trail, spend control

5.2 Components and interfaces

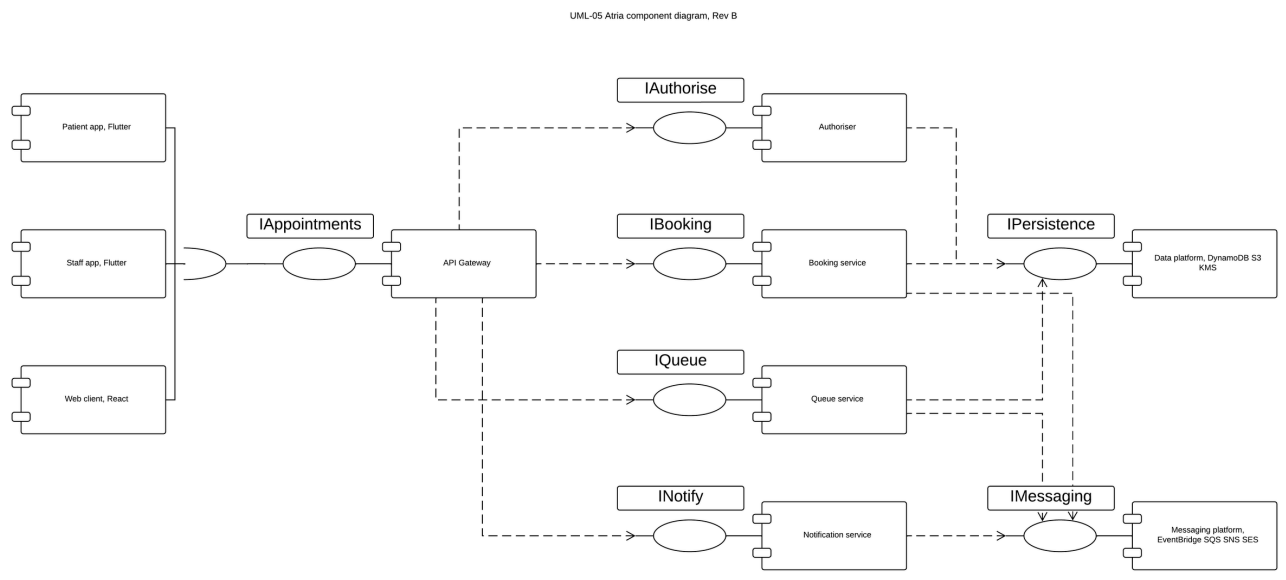


Figure 9. UML-05 component diagram. Clients require IAppointments from API Gateway, which depends on the authoriser and the booking, queue and notification services; services depend on the persistence and messaging platforms.

COMPONENT	INTERFACE	RESPONSIBILITY
Authoriser	IAuthorise	Verifies the Cognito JWT against cached keys, resolves tenant, person and roles, applies the permission matrix and the guard rules, and returns an allow or deny policy with context.
Booking service	IBooking	Specialist bookings, procedure requests and placement, reschedule and cancel with entitlement, disruption offers, care context, referrals.
Queue service	IQueue	Sessions, queue tickets with server-assigned order, live arrival windows, walk-ins, check-in and next-patient assignment.
Staff and identity	IAppointments	Profile completion, phone verification, guardian link, patient register, staff provisioning with roles, priority allowances.

COMPONENT	INTERFACE	RESPONSIBILITY
Tenant configuration	IAppointments	Clinics, appointment types, intake forms, fee bands, region pack, feature flags.
Fee ledger and quality	IAppointments	Ledger entries at closure, feedback collection and role-scoped feedback reads, reports.
Notification service	INotify	Consumes appointment events from the outbox, composes notices from templates in the patient's language and hands them to the reminder worker.
Reminder worker	IMessaging	Consumes the reminder queue, re-reads the appointment and current contact details at send time, sends through SNS or SES and records the message log.

5.3 Infrastructure as code

All infrastructure is AWS CDK v2 in Python, one stack per concern. The region and account are per-environment parameters, so development and production need not share a region. Everything that runs on AWS is Python, including the CDK application and every Lambda handler, so the specification modules that define the roles, the permission matrix, the data model, the key design and the lifecycles are imported by the code that enforces them rather than restated in a second language.

Lambda packaging has two assets: the dependency layer, published once by the platform stack, and the handler code. The layer is installed for the Lambda runtime and architecture rather than for the machine that builds it, so a compiled wheel arrives as the Linux build the runtime needs and a deployment from a developer's Windows machine is identical to one from the pipeline. No container runtime is needed to synthesise or deploy.

STACK	CONTENTS
`NetworkStack`	CloudFront distribution, the production AWS WAF web ACL, certificates and DNS records. Lambda functions run outside a VPC; every dependency is a regional AWS endpoint reached over IAM-signed calls.
`DataStack`	DynamoDB main table with its indexes, the care context table on its own customer-managed key, the S3 bucket for referral documents, KMS keys, point-in-time recovery and backups.
`PlatformStack`	The Lambda dependency layer every function shares, published once. It sits below the stacks that use it because two of them need the same layer and a stack cannot add a trigger to another stack's user pool.
`IdentityStack`	Cognito user pool, patient, staff and admin app clients, Google identity provider on the patient client only, custom attributes, and the pre-token generation trigger that adds the tenant, person and role claims. The trigger is the V2 form: only that version can add claims to an access token, and the API authorises on the access token.
`ApiStack`	API Gateway REST API, request validators, usage plans, the Lambda authoriser and the service functions (booking, queue, staff and identity, tenant configuration, fee ledger and quality).
`AsyncStack`	EventBridge bus and Scheduler, SQS main and FIFO outbox queues with dead letter queues, the notification service and reminder worker, SNS and SES configuration.
`ObservabilityStack`	Log groups with explicit retention, metric filters, alarms, one dashboard per environment, X-Ray tracing and CloudTrail.

STACK	CONTENTS
`CostGuardStack`	AWS Budgets at 50, 80 and 100 per cent with SNS and email alerts, Cost Anomaly Detection and the per-tenant daily SMS spend cap.

5.4 Environments and region

ENVIRONMENT	REGION	DATA	DEPLOYS FROM
Development	us-east-1	Synthetic only, generated from the Cameroon region pack	Every merge to the integration branch
Staging	Production region	Synthetic only	Tagged release after manual approval
Production	Decided before Phase 3 (OI-02)	Real patient data only once authorisations are held	Tagged release after manual approval

No real patient data outside an authorised production region.
Any AWS region is a transfer of personal data abroad under Law No. 2024/017. Development is defensible only because it holds no real personal data. Seed, demo and test data are generated from the region pack and are never copied from a live system.

5.5 Asynchronous processing

- Every write that other parts of the system react to also writes an outbox record in the same transaction. A stream consumer publishes outbox records to a FIFO SQS queue keyed on appointment id, because ordering per appointment is the one place it matters.
- EventBridge delivers into SQS rather than directly into a function, so a poison message is parked in the dead letter queue and can be redriven instead of being retried into nothing.
- A reminder is a one-time EventBridge Scheduler schedule created at booking. When it fires, the reminder worker re-reads the appointment and the patient's current contact details; a cancelled or moved appointment sends nothing.
- Queue movement recalculates arrival windows for everyone after the current position and sends a queue-moved SMS only when a window changes by more than the tenant's threshold.
- Delivery receipts from SNS and delivery, bounce and complaint events from SES update the message log.

5.6 Security

CONTROL	IMPLEMENTATION
Transport	TLS 1.2 or later everywhere; HSTS on the web app; WAF managed rule sets and rate-based rules on the API.
Identity	Cognito with separate patient, staff and admin app clients. Self-registration exists only on the patient client. Admin sign-in requires a second factor.
Authorisation	A Lambda authoriser applies the role and scope matrix and seven guard rules on every request (section 8). Handlers re-check ownership against the resolved person.

CONTROL	IMPLEMENTATION
Encryption at rest	Customer-managed KMS keys. Care context lives in its own table on its own key, and only the booking and queue services' roles may use that key.
Secrets	Third-party keys, including the Google client secret and any SMS or voice provider key, live in Secrets Manager. None appears in code, templates or environment variables.
Logging	Structured JSON logs with a correlation id; phone numbers, email addresses and names are masked. CloudTrail records every control-plane action.
Offline clients	The patient app shows its last sync offline and refuses bookings without a connection. The staff app queues arrival and no-show marks and replays them; a mark on an appointment cancelled meanwhile is rejected and shown.

6 Data model

26	225	38	5
Entities	Attributes	Relationships	Packages

6.1 Class model

The class model is split into four packages so each sheet can be read at print size.

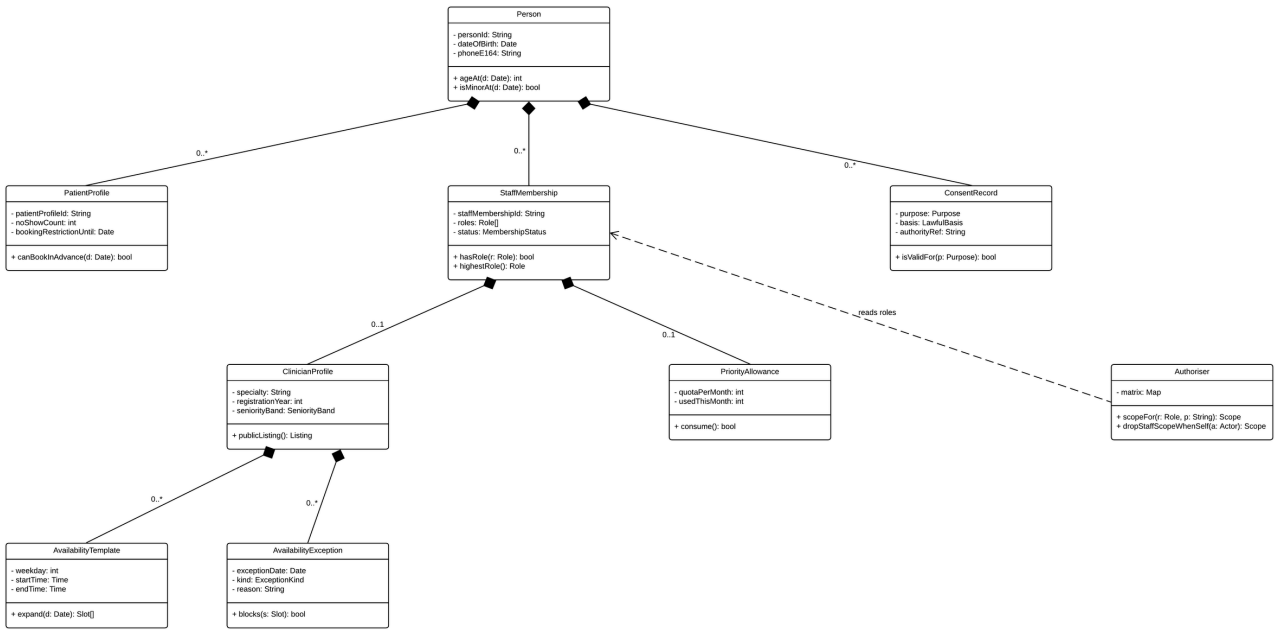


Figure 10. UML-02a identity and access. Person generalises into patient profile and staff membership; clinician profile, consent and priority allowance.

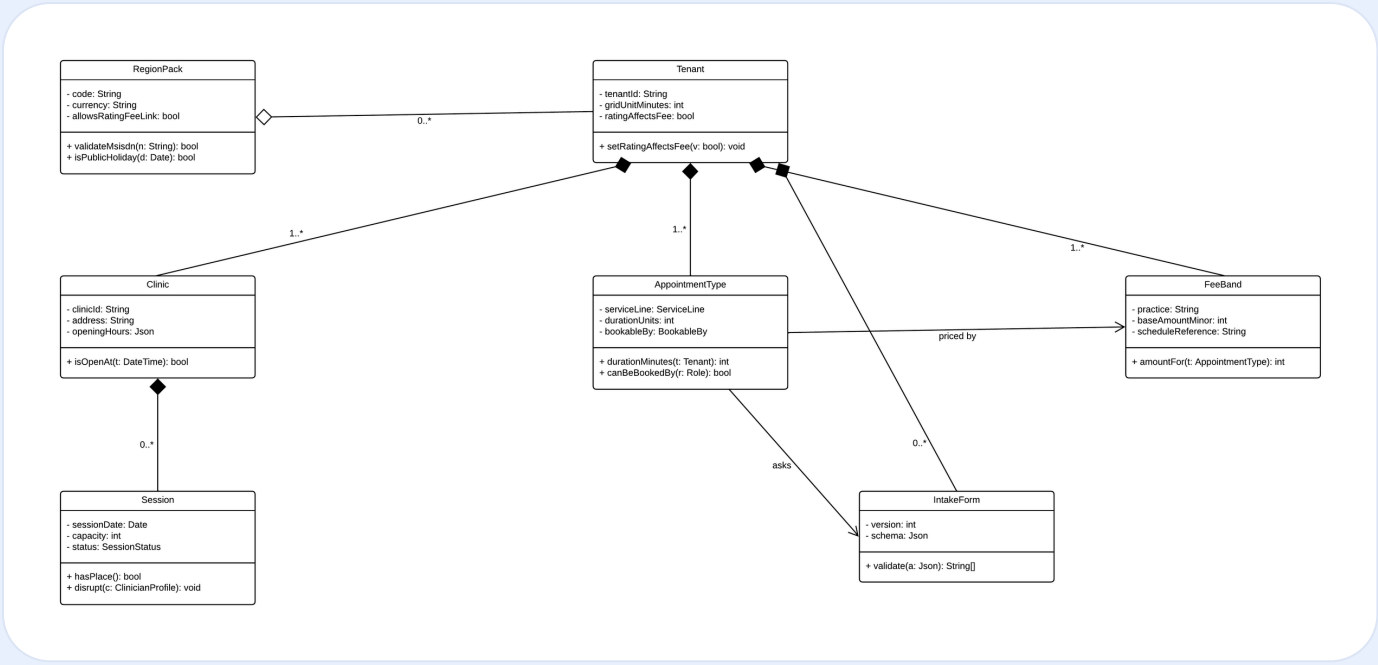


Figure 11. UML-02b tenancy and configuration. Region pack, tenant, clinic, appointment type, intake form and fee band.

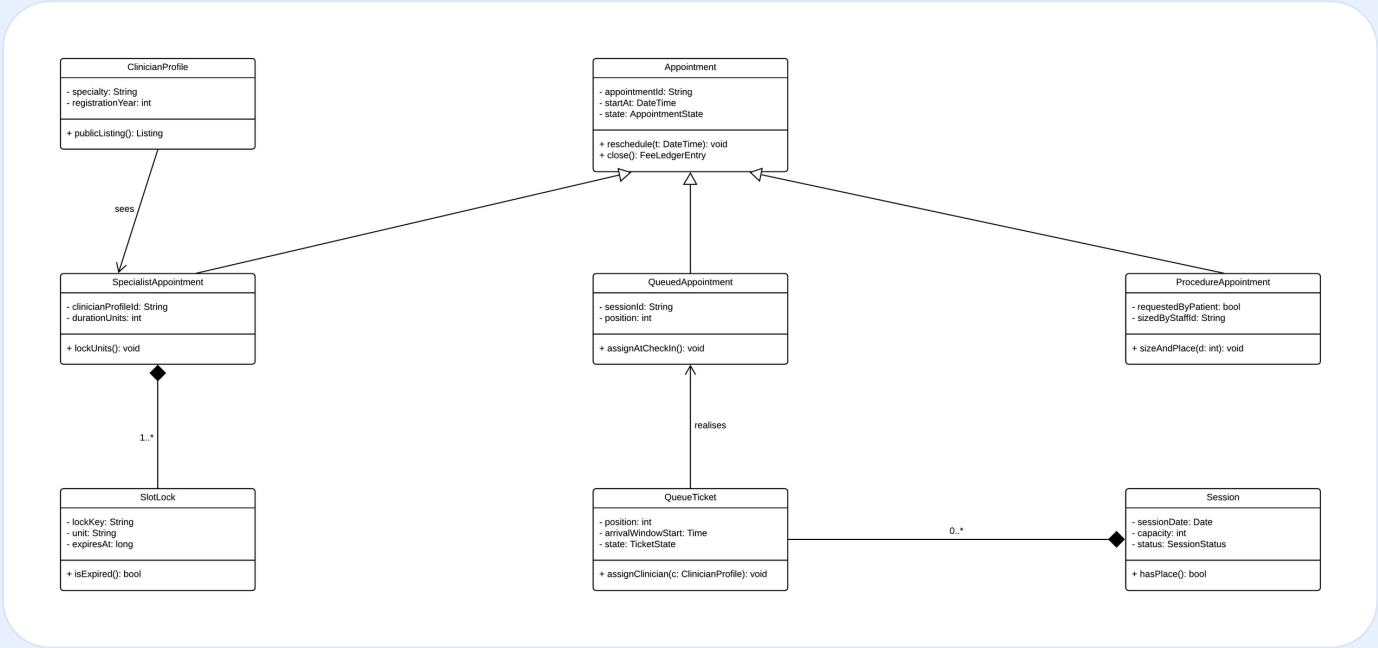


Figure 12. UML-02c appointment specialisation. Specialist, general and procedure appointments with their session, ticket and slot lock.

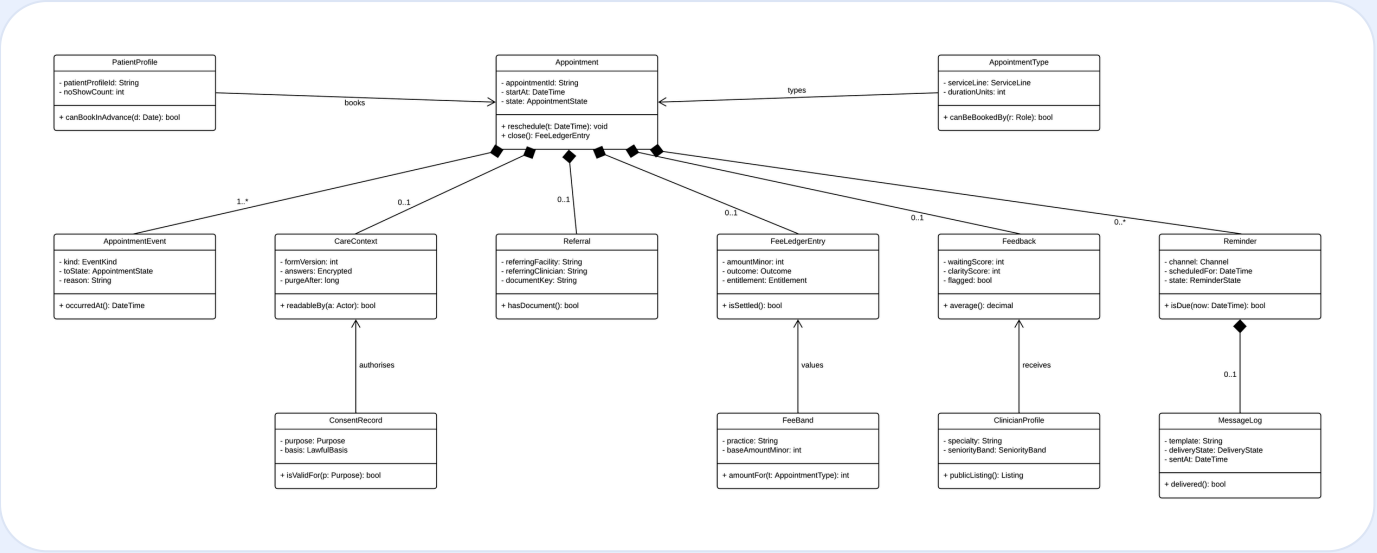


Figure 13. UML-02d appointment lifecycle records. Events, referral, care context, fee ledger entry, feedback, reminder and message log.

6.2 Entity relationships

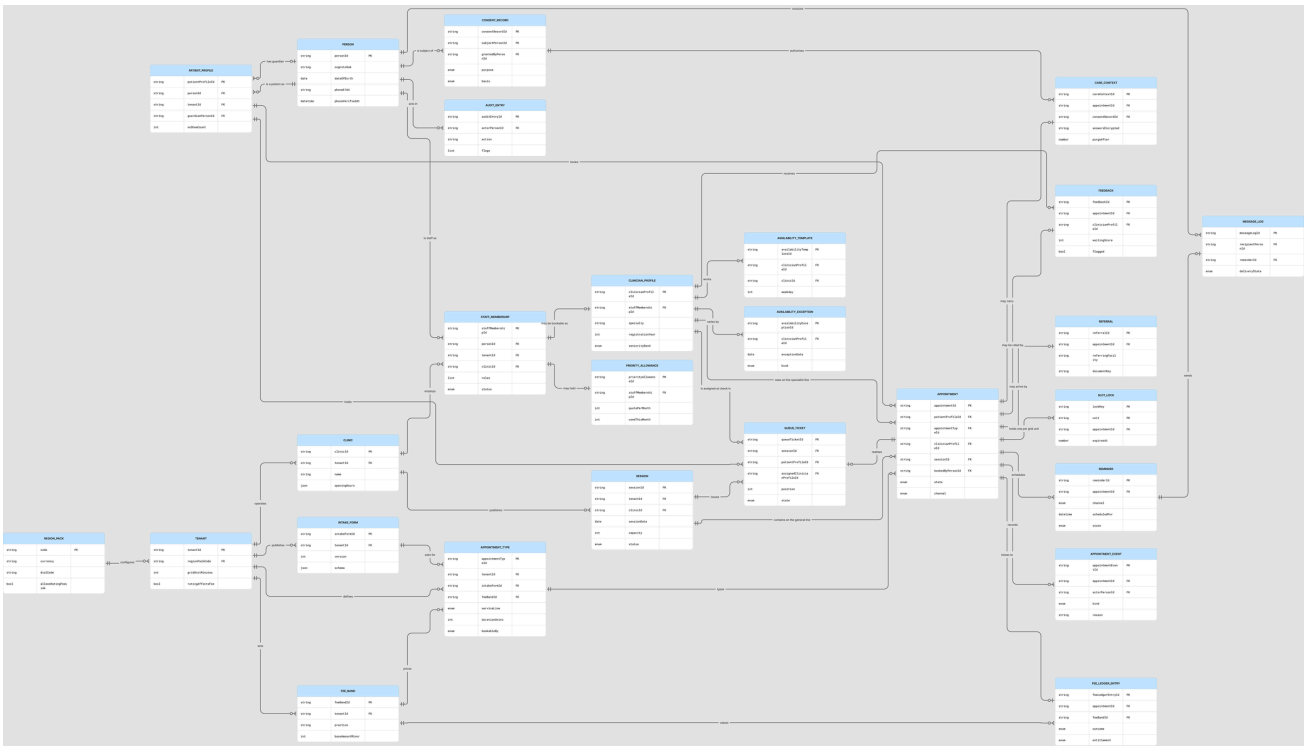


Figure 14. UML-08 entity relationship diagram in crow's foot notation, all entities and relationships.

FROM	CARDINALITY	TO	MEANING
REGION_PACK	one to zero or more	**TENANT**	configures
TENANT	one to zero or more	**CLINIC**	operates
TENANT	one to zero or more	**APPOINTMENT_TYPE**	defines

FROM	CARDINALITY	TO	MEANING
TENANT	one to zero or more	FEE_BAND	sets
TENANT	one to zero or more	INTAKE_FORM	publishes
FEE_BAND	one to zero or more	APPOINTMENT_TYPE	prices
INTAKE_FORM	one to zero or more	APPOINTMENT_TYPE	asks for
PERSON	one to zero or more	PATIENT_PROFILE	is a patient as
PERSON	one to zero or more	STAFF_MEMBERSHIP	is staff as
PERSON	one to zero or more	CONSENT_RECORD	is subject of
PATIENT_PROFILE	zero or more to zero or one	PERSON	has guardian
STAFF_MEMBERSHIP	one to zero or one	CLINICIAN_PROFILE	may be bookable as
STAFF_MEMBERSHIP	one to zero or one	PRIORITY_ALLOWANCE	may hold
CLINIC	one to zero or more	STAFF_MEMBERSHIP	employs
CLINICIAN_PROFILE	one to zero or more	AVAILABILITY_TEMPLATE	works
CLINICIAN_PROFILE	one to zero or more	AVAILABILITY_EXCEPTION	varies by
CLINIC	one to zero or more	SESSION	publishes
SESSION	one to zero or more	QUEUE_TICKET	issues
PATIENT_PROFILE	one to zero or more	QUEUE_TICKET	holds
CLINICIAN_PROFILE	one to zero or more	QUEUE_TICKET	is assigned at check in
PATIENT_PROFILE	one to zero or more	APPOINTMENT	books
APPOINTMENT_TYPE	one to zero or more	APPOINTMENT	types
CLINICIAN_PROFILE	one to zero or more	APPOINTMENT	sees on the specialist line
SESSION	one to zero or more	APPOINTMENT	contains on the general line
QUEUE_TICKET	zero or one to exactly one	APPOINTMENT	realises
APPOINTMENT	one to zero or more	SLOT_LOCK	holds one per grid unit
APPOINTMENT	one to zero or more	APPOINTMENT_EVENT	records
APPOINTMENT	one to zero or one	REFERRAL	may arrive by
APPOINTMENT	one to zero or one	CARE_CONTEXT	may carry
CONSENT_RECORD	one to zero or more	CARE_CONTEXT	authorises

FROM	CARDINALITY	TO	MEANING
APPOINTMENT	one to zero or one	FEE_LEDGER_ENTRY	closes to
FEE_BAND	one to zero or more	FEE_LEDGER_ENTRY	values
APPOINTMENT	one to zero or one	FEEDBACK	may be rated by
CLINICIAN_PROFILE	one to zero or more	FEEDBACK	receives
APPOINTMENT	one to zero or more	REMINDER	schedules
REMINDER	one to zero or one	MESSAGE_LOG	sends
PERSON	one to zero or more	MESSAGE_LOG	receives
PERSON	one to zero or more	AUDIT_ENTRY	acts in

6.3 Entity catalogue

TENANCY AND CONFIGURATION

REGION_PACK. The locale a tenant operates in. Supplies everything country specific so a second country is a data change, not a code change. AD-13.

ATTRIBUTE	TYPE	KEY	NOTE
code	string	PK	ISO country code, CM for Cameroon
currency	string		XAF
dialCode	string		+237
msisdnPattern	string		Validation for mobile numbers
languages	list		fr, en
timezone	string		Africa/Douala
administrativeRegions	list		The ten regions
nationalIdPattern	string		CNI format
publicHolidays	list		Used to suppress booking and reminders
emergencyNumber	string		Shown on disruption notices
smsSenderIdPolicy	string		Registration rules per network
allowsRatingFeeLink	bool		False for CM. Enforces AD-11 at the data layer

TENANT. An institution using Atria. Every record below belongs to exactly one.

ATTRIBUTE	TYPE	KEY	NOTE
tenantId	string	PK	
name	string		
regionPackCode	string	FK	REGION_PACK
gridUnitMinutes	int		Calendar resolution, default 10. AD-04
qualityProgrammeEnabled	bool		Master switch for feedback collection. AD-11
ratingAffectsFee	bool		Refused when the region pack forbids it
noShowStrikeThreshold	int		0 disables strikes, which is the default. AD-07
noShowStrikeWindowDays	int		Rolling window, default 180
status	enum		ONBOARDING, ACTIVE, SUSPENDED

CLINIC. A physical site within a tenant where appointments take place.

ATTRIBUTE	TYPE	KEY	NOTE
clinicId	string	PK	
tenantId	string	FK	TENANT
name	string		
address	string		Formatted per the region pack
phone	string		
openingHours	json		Per weekday

APPOINTMENT_TYPE. What is being booked. Carries its own duration, which is the change that removes the single tenant wide slot length. AD-04.

ATTRIBUTE	TYPE	KEY	NOTE
appointmentTypeId	string	PK	
tenantId	string	FK	TENANT
code	string		GEN_CONSULT, SPEC_FIRST, MINOR_PROC
name	string		
serviceLine	enum		GENERAL, SPECIALIST, PROCEDURE. AD-01
durationUnits	int		In grid units. Null for queued general consultations

ATTRIBUTE	TYPE	KEY	NOTE
bufferUnits	int		Turnaround held after the appointment
bookableBy	enum		PATIENT, STAFF, BOTH. Procedures are STAFF. AD-05
patientRequestable	bool		Lets a patient raise a REQUESTED procedure. AD-05
minNoticeMinutes	int		
maxAdvanceDays	int		
cancellationWindowMinutes	int		Boundary between in window and late. AD-07
intakeFormId	string	FK	INTAKE_FORM, nullable
feeBandId	string	FK	FEE_BAND
active	bool		

INTAKE_FORM. The questions asked at booking, chosen by appointment type. AD-06.

ATTRIBUTE	TYPE	KEY	NOTE
intakeFormId	string	PK	
tenantId	string	FK	TENANT
name	string		
version	int		Answers pin the version they were captured against
schema	json		Field definitions, types, required flags, translations

FEE_BAND. The baseline fee for a practice, not for an individual. Set by the tenant administrator from the ministerial schedule. AD-11.

ATTRIBUTE	TYPE	KEY	NOTE
feeBandId	string	PK	
tenantId	string	FK	TENANT
practice	string		General practice, cardiology, paediatrics
baseAmountMinor	int		Minor units of the region pack currency
scheduleReference	string		The order the amount is taken from
effectiveFrom	date		

PERSON. One human being. Holds no role. This split is what lets a staff member be a patient without being able to act on their own appointment as staff. AD-10.

ATTRIBUTE	TYPE	KEY	NOTE
`personId`	string	PK	
`cognitoSub`	string		Identity provider subject
`givenName`	string		
`familyName`	string		
`dateOfBirth`	date		Stored as a date. Age is derived, never stored. AD-09
`phoneE164`	string		Reminder channel
`phoneVerifiedAt`	datetime		Set only by one time code, whatever the source
`email`	string		
`preferredLanguage`	string		From the region pack's language list

PATIENT_PROFILE. A person as a patient of one tenant.

ATTRIBUTE	TYPE	KEY	NOTE
`patientProfileId`	string	PK	
`personId`	string	FK	PERSON
`tenantId`	string	FK	TENANT
`patientNumber`	string		The clinic's own identifier
`guardianPersonId`	string	FK	PERSON, required while under 18. AD-09
`notificationPrefs`	json		Channel, lead times, proxy recipient
`noShowCount`	int		Rolling count in the tenant's window
`bookingRestrictionUntil`	date		Advance booking withheld until this date. AD-07

STAFF_MEMBERSHIP. A person's employment at one clinic, carrying their roles.

ATTRIBUTE	TYPE	KEY	NOTE
`staffMembershipId`	string	PK	
`personId`	string	FK	PERSON

ATTRIBUTE	TYPE	KEY	NOTE
tenantId	string	FK	TENANT
clinicId	string	FK	CLINIC
roles	list		RECEPTIONIST, CLINICIAN, CLINIC_MANAGER, TENANT_ADMIN
employeeNumber	string		
status	enum		INVITED, ACTIVE, SUSPENDED, ENDED

CLINICIAN_PROFILE. The bookable part of a staff membership. Only the facts Article 7 permits in a listing appear on the patient side. AD-11.

ATTRIBUTE	TYPE	KEY	NOTE
clinicianProfileId	string	PK	
staffMembershipId	string	FK	STAFF_MEMBERSHIP
specialty	string		
qualifications	list		Officially recognised titles only
registrationYear	int		Year entered on the Order's roll
ordreNumber	string		Registration number
languages	list		
seniorityBand	enum		Derived from registrationYear. Sorts the directory without asserting a quality claim

CONSENT_RECORD. Who consented to what, when and how. Required by Law 2024/017 and the evidence a pilot has to produce.

ATTRIBUTE	TYPE	KEY	NOTE
consentRecordId	string	PK	
subjectPersonId	string	FK	PERSON, whose data it is
grantedByPersonId	string	FK	PERSON, the guardian where the subject is a minor
purpose	enum		BOOKING, REMINDERS, CARE_CONTEXT, FEEDBACK
basis	enum		CONSENT, LEGAL_OBLIGATION, PUBLIC_INTEREST, VITAL_INTEREST
method	enum		APP, WEB, DESK_SIGNATURE, SMS_REPLY
authorityAuthorisationRef	string		Reference for a minor's sensitive data. OI-11

ATTRIBUTE	TYPE	KEY	NOTE
grantedAt	datetime		
withdrawnAt	datetime		

PRIORITY_ALLOWANCE. An explicit, bounded, visible staff booking privilege. The alternative to a loophole. AD-10 rule 5.

ATTRIBUTE	TYPE	KEY	NOTE
priorityAllowanceId	string	PK	
staffMembershipId	string	FK	STAFF_MEMBERSHIP
quotaPerMonth	int		
usedThisMonth	int		
grantedByPersonId	string	FK	PERSON

SCHEDULING

AVAILABILITY_TEMPLATE. A clinician's recurring working hours.

ATTRIBUTE	TYPE	KEY	NOTE
availabilityTemplateId	string	PK	
clinicianProfileId	string	FK	CLINICIAN_PROFILE
clinicId	string	FK	CLINIC
weekday	int		0 to 6
startTime	time		
endTime	time		
effectiveFrom	date		
effectiveTo	date		

AVAILABILITY_EXCEPTION. A one off change. Marking a clinician unavailable here is what starts a disruption. AD-03.

ATTRIBUTE	TYPE	KEY	NOTE
availabilityExceptionId	string	PK	
clinicianProfileId	string	FK	CLINICIAN_PROFILE

ATTRIBUTE	TYPE	KEY	NOTE
exceptionDate	date		
kind	enum		UNAVAILABLE, EXTRA
startTime	time		
endTime	time		
reason	string		Recorded, and shown to affected patients in general terms

SESSION. A published block of general consultation capacity. Patients book a place in it rather than a named doctor. AD-02.

ATTRIBUTE	TYPE	KEY	NOTE
sessionId	string	PK	
tenantId	string	FK	TENANT
clinicId	string	FK	CLINIC
sessionDate	date		
startTime	time		
endTime	time		
rosteredClinicianCount	int		
expectedThroughputPerHour	decimal		Seeded, then corrected from measurement
capacity	int		Derived, and the number the queue is capped at
issuedTickets	int		
actualThroughputPerHour	decimal		Written at close. Feeds next week's estimate
status	enum		PLANNED, OPEN, RUNNING, DISRUPTED, CLOSED

QUEUE_TICKET. A patient's place in a session. Position is assigned by the server from the request timestamp, which is AD-10 rule 2.

ATTRIBUTE	TYPE	KEY	NOTE
queueTicketId	string	PK	
sessionId	string	FK	SESSION
patientProfileId	string	FK	PATIENT_PROFILE

ATTRIBUTE	TYPE	KEY	NOTE
position	int		Server assigned, never client supplied
arrivalWindowStart	time		Recalculated live as the session runs
arrivalWindowEnd	time		
state	enum		ISSUED, NOTIFIED, ARRIVED, IN_CONSULTATION, COMPLETED, NO_SHOW, WITHDRAWN
assignedClinicianProfileId	string	FK	CLINICIAN_PROFILE, set at check in
appointmentId	string	FK	APPOINTMENT

SLOT_LOCK. The item whose existence prevents double booking. One per grid unit, all written in a single transaction so an N unit appointment is still atomic. AD-04.

ATTRIBUTE	TYPE	KEY	NOTE
lockKey	string	PK	tenantId, clinicianProfileId and date
unit	string	SK	Start of the grid unit, HHMM
appointmentId	string	FK	APPOINTMENT
expiresAt	number		TTL, releases an abandoned hold

APPOINTMENT. The central record. Holds either a clinician and a time, or a session and a ticket, never both.

ATTRIBUTE	TYPE	KEY	NOTE
appointmentId	string	PK	
tenantId	string	FK	TENANT
clinicId	string	FK	CLINIC
patientProfileId	string	FK	PATIENT_PROFILE
appointmentTypeId	string	FK	APPOINTMENT_TYPE
clinicianProfileId	string	FK	CLINICIAN_PROFILE, null on the general line
sessionId	string	FK	SESSION, null on the specialist line
startAt	datetime		
endAt	datetime		Derived from the type's durationUnits
state	enum		See the state machine

ATTRIBUTE	TYPE	KEY	NOTE
channel	enum		ONLINE, WALK_IN, PHONE, REFERRAL. AD-12
bookedByPersonId	string	FK	PERSON
bookedByRole	enum		PATIENT or the staff role used. Blank means self service
referralId	string	FK	REFERRAL, nullable
careContextId	string	FK	CARE_CONTEXT, nullable
version	int		Optimistic concurrency for offline staff writes

APPOINTMENT_EVENT. Every transition, with who caused it and why. Drives the patient timeline and satisfies the audit obligation.

ATTRIBUTE	TYPE	KEY	NOTE
appointmentEventId	string	PK	
appointmentId	string	FK	APPOINTMENT
kind	enum		BOOKED, CONFIRMED, RESCHEDULED, DISRUPTED, CANCELLED, ARRIVED, COMPLETED, NO_SHOW
fromState	enum		
toState	enum		
actorPersonId	string	FK	PERSON
actorRole	enum		
reason	string		Required on every staff initiated reschedule. AD-10 rule 4
occurredAt	datetime		

REFERRAL. Where a specialist appointment came from. The normal door into the specialist line, matching the health pyramid. AD-12.

ATTRIBUTE	TYPE	KEY	NOTE
referralId	string	PK	
appointmentId	string	FK	APPOINTMENT
referringFacility	string		
referringClinicianName	string		
referralNote	string		

ATTRIBUTE	TYPE	KEY	NOTE
documentKey	string		Object store key, encrypted at rest
receivedAt	datetime		

CARE_CONTEXT. What the patient chose to send ahead of one visit. Deliberately its own entity, its own key and its own shorter retention. AD-06.

ATTRIBUTE	TYPE	KEY	NOTE
careContextId	string	PK	
appointmentId	string	FK	APPOINTMENT
intakeFormId	string	FK	INTAKE_FORM
formVersion	int		
answers	encrypted		Readable by the assigned clinician and the patient only
consentRecordId	string	FK	CONSENT_RECORD
submittedAt	datetime		
purgeAfter	number		TTL. Shorter than the booking history. AD-08

ACCOUNTABILITY AND QUALITY

FEE_LEDGER_ENTRY. What was owed and what the outcome entitles the patient to. Nothing settles until a payment provider arrives in Phase 3. AD-07.

ATTRIBUTE	TYPE	KEY	NOTE
feeLedgerEntryId	string	PK	
appointmentId	string	FK	APPOINTMENT
tenantId	string	FK	TENANT
feeBandId	string	FK	FEE_BAND
amountMinor	int		
currency	string		
outcome	enum		ATTENDED, CANCELLED_IN_WINDOW, CANCELLED_LATE, NO_SHOW, CLINIC_CANCELLED, CLINIC_RESCHEDULED
entitlement	enum		NONE, PARTIAL, FULL
settledAt	datetime		Null throughout Phase 2

FEEDBACK. Collected after the scheduled end, only where attendance was recorded. Seen by the tenant administrator. Never shown on a patient facing directory. AD-11.

ATTRIBUTE	TYPE	KEY	NOTE
feedbackId	string	PK	
appointmentId	string	FK	APPOINTMENT
patientProfileId	string	FK	PATIENT_PROFILE
clinicianProfileId	string	FK	CLINICIAN_PROFILE
waitingScore	int		1 to 5
clarityScore	int		
respectScore	int		
wouldReturn	bool		
comment	string		
flagged	bool		Safety or conduct concern, escalated to the clinic manager
submittedAt	datetime		

MESSAGING AND AUDIT

REMINDER. One scheduled send. Queued through SQS so a failure is parked and redrivable rather than lost. AD-14.

ATTRIBUTE	TYPE	KEY	NOTE
reminderId	string	PK	
appointmentId	string	FK	APPOINTMENT
channel	enum		SMS, EMAIL, VOICE
leadMinutes	int		
scheduledFor	datetime		
state	enum		SCHEDULED, SENT, DELIVERED, FAILED, CANCELLED
attempts	int		
providerMessageId	string		

MESSAGE_LOG. What was actually sent, to whom, and what the provider said.

ATTRIBUTE	TYPE	KEY	NOTE
messageLogId	string	PK	
tenantId	string	FK	TENANT
recipientPersonId	string	FK	PERSON
reminderId	string	FK	REMINDER, nullable
channel	enum		
template	string		
deliveryState	enum		
sentAt	datetime		

AUDIT_ENTRY. Every privileged action. A legal obligation here, not a convenience.

ATTRIBUTE	TYPE	KEY	NOTE
auditEntryId	string	PK	
tenantId	string	FK	TENANT
actorPersonId	string	FK	PERSON
actorRole	enum		
action	string		
subjectType	string		
subjectId	string		
reason	string		
flags	list		POTENTIAL_SELF_BENEFIT raised by AD-10 rule 3
occurredAt	datetime		

6.4 Storage and keys

The main store is one DynamoDB table with tenant-prefixed partition keys, so no query can cross a tenant boundary by construction. Care context is the one exception: it lives in its own table, encrypted under its own key, with a time to live that purges it on the shorter clock.

ENTITY	PARTITION KEY	SORT KEY	TABLE
Tenant	TENANT#t	META	Main

ENTITY	PARTITION KEY	SORT KEY	TABLE
Region pack	`REGION#code`	`PACK`	Main
Clinic	`TENANT#t#CLINIC#c`	`PROFILE`	Main
Appointment type	`TENANT#t#TYPE#id`	`TYPE`	Main
Intake form	`TENANT#t#FORM#id`	`V#version`	Main
Fee band	`TENANT#t#FEE#id`	`FROM#effectiveFrom`	Main
Person	`PERSON#p`	`PERSON`	Main
Patient profile	`TENANT#t#PAT#pp`	`PROFILE`	Main
Staff membership	`TENANT#t#STAFF#s`	`MEMBERSHIP`	Main
Clinician profile	`TENANT#t#STAFF#s`	`CLINICIAN`	Main
Consent record	`PERSON#p`	`CONSENT#grantedAt#id`	Main
Priority allowance	`TENANT#t#STAFF#s`	`PRIORITY`	Main
Availability template	`TENANT#t#CLIN#c`	`AVAIL#weekday`	Main
Availability exception	`TENANT#t#CLIN#c`	`EXC#start`	Main
Session	`TENANT#t#SESSION#id`	`SESSION`	Main
Queue ticket	`TENANT#t#SESSION#id`	`TICKET#requestTs#ticketId`	Main
Slot lock	`TENANT#t#LOCK#clinicianId#unitStart`	`LOCK`	Main
Appointment	`TENANT#t#APPT#id`	`APPT`	Main
Appointment event	`TENANT#t#APPT#id`	`EVENT#ts`	Main
Referral	`TENANT#t#APPT#id`	`REFERRAL`	Main
Fee ledger entry	`TENANT#t#APPT#id`	`LEDGER`	Main
Feedback	`TENANT#t#APPT#id`	`FEEDBACK`	Main
Reminder	`TENANT#t#APPT#id`	`REMINDER#sendAt`	Main
Message log	`TENANT#t#MSG#yyyy-mm-dd`	`sentAt#messageId`	Main
Audit entry	`TENANT#t#AUDIT#yyyy-mm-dd`	`ts#auditId`	Main
Care context	`TENANT#t#APPT#id`	`CARE`	Care context table, own key, TTL

INDEX	PARTITION KEY	SORT KEY	SERVES
PatientIndex	`patientProfileId`	`startAt`	A patient's own appointments and tickets, all four range views
ClinicianIndex	`clinicianProfileId`	`startAt`	A clinician's calendar
ClinicDayIndex	`clinicId#date`	`startAt`	The clinic day, the week and the requests inbox
PersonIndex	`cognitoSub`	`none`	Resolving the signed-in person on every request
OutboxIndex	`outboxShard`	`createdAt`	Change capture into the FIFO outbox, keyed on appointment id
DirectoryIndex	`directoryKey`	`directorySort`	The bookable clinicians of one tenant, most senior first

Double booking stays impossible with typed durations.
A specialist booking of N grid units writes the appointment and N slot locks in one TransactWriteItems call, each lock with a condition that it does not already exist. If any unit is taken the whole transaction fails with 409 and nothing is written.

6.5 Retention

DATA	RETAINED	SHOWN OR KEPT LIVE
Appointments, events, ledger	Ten years (proposal, OI-12)	Five years to the patient by default, tenant-configurable
Care context	Purged after the visit's review window, default 90 days	Patient and assigned clinician only
Message log	12 months	Staff message log
Audit entries	Ten years	Read-only to clinic manager (clinic) and tenant administrator (tenant)
Consent records	Life of the account plus five years	Patient settings
Slot locks	Removed with the slot	Not shown

7 Behaviour

7.1 Sequences

STAFF ACCOUNT CREATION AND FIRST SIGN-IN

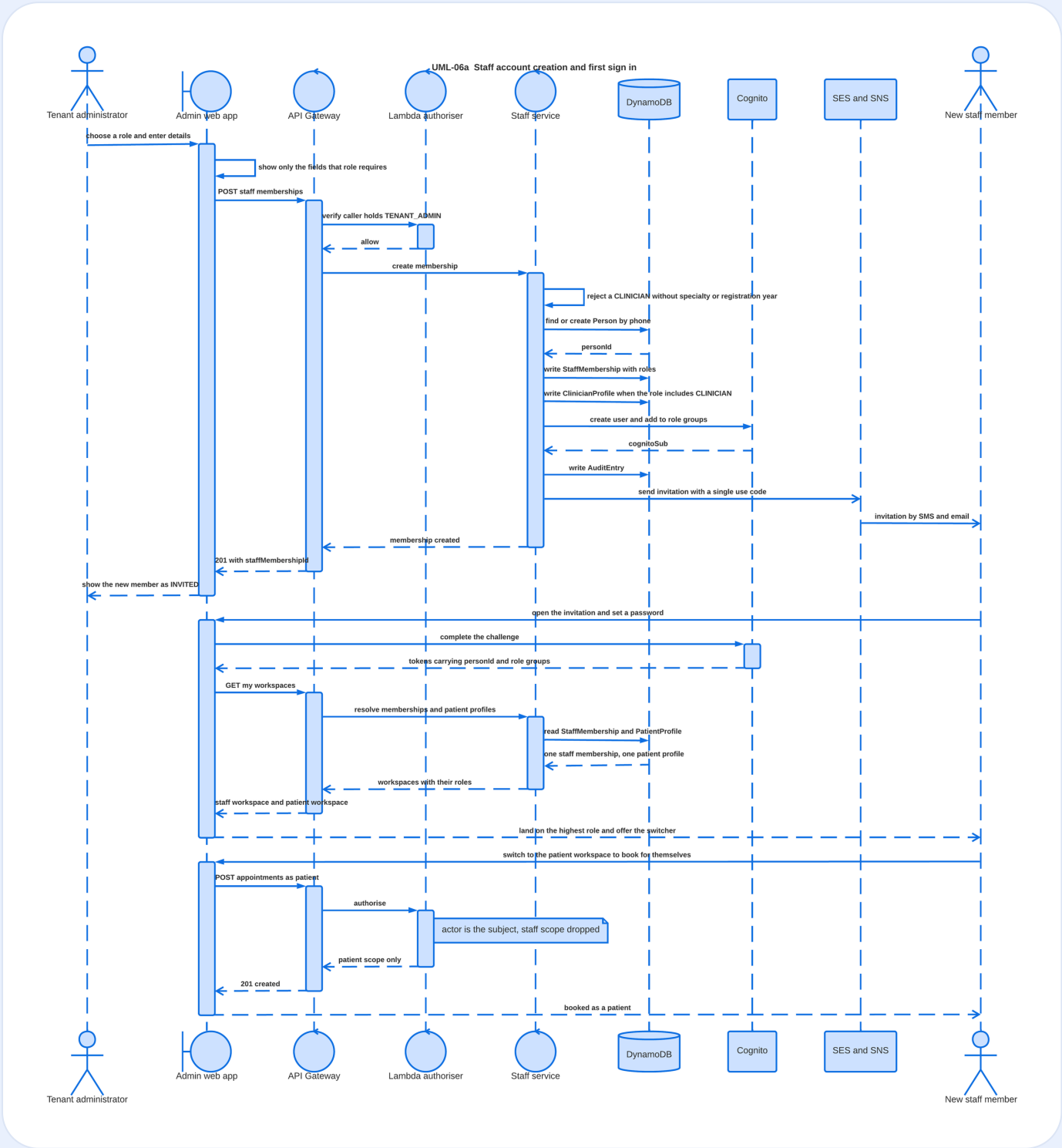


Figure 15. UML-06a staff account creation and first sign-in.

- The tenant administrator submits the staff member's details and roles. The service enforces the fields required for each role and creates the Cognito user with a temporary password and a staff membership carrying the roles.

- The staff member signs in with the temporary password and must set a new one. The token carries the tenant, person and roles, and the authoriser grants exactly those roles' permissions.
- Every step writes an audit entry.

SPECIALIST BOOKING WITH A TYPED DURATION

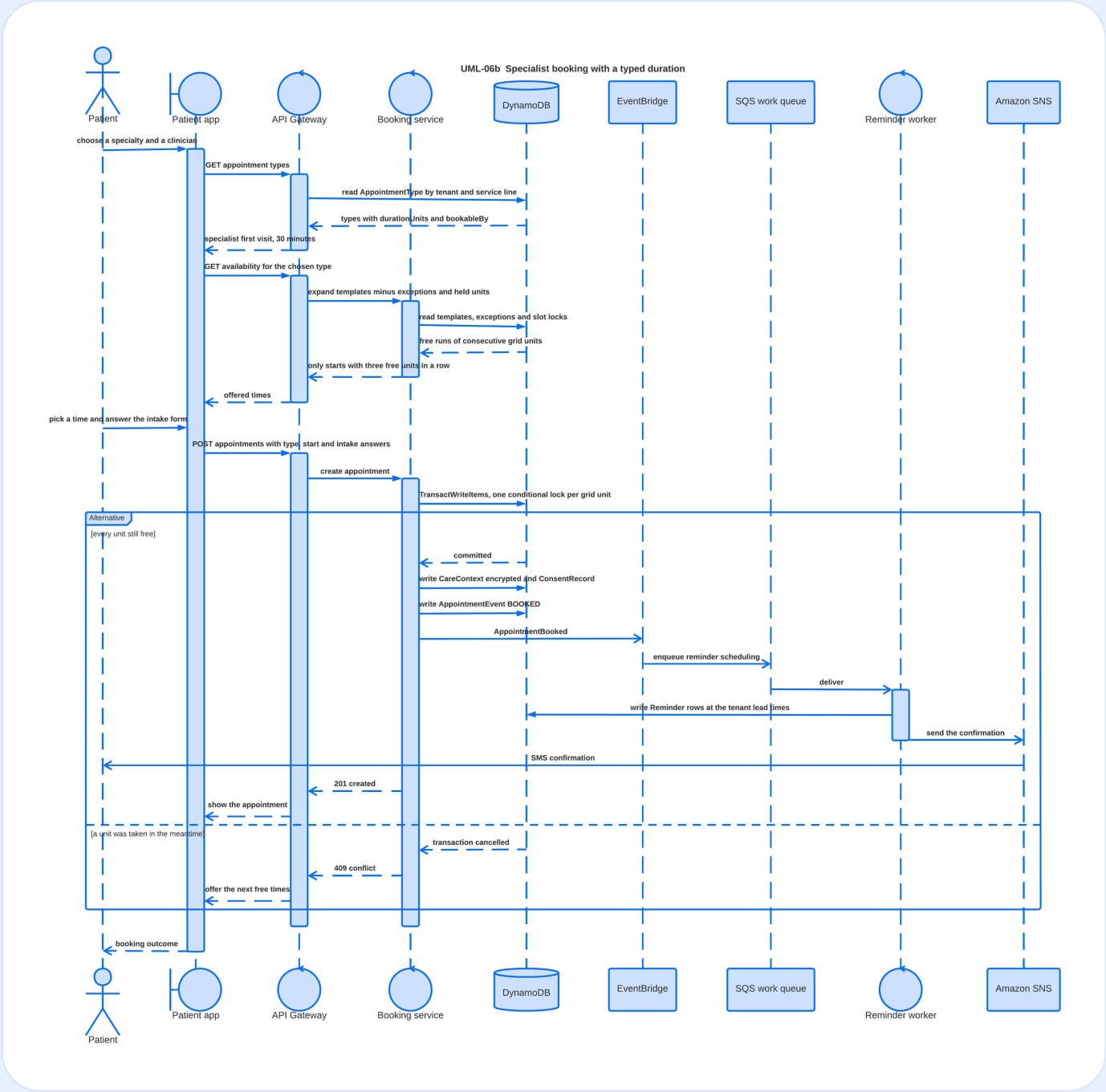


Figure 16. UML-06b specialist booking with a typed duration.

- The client lists free start units for the chosen type; a start is free only when every unit of duration plus buffer is free.
- The booking writes the appointment, N slot locks, the care context and the outbox record in one transaction. A conflict returns 409 and the client offers the nearest free starts.
- The outbox drives the confirmation email and SMS and the reminder schedule.

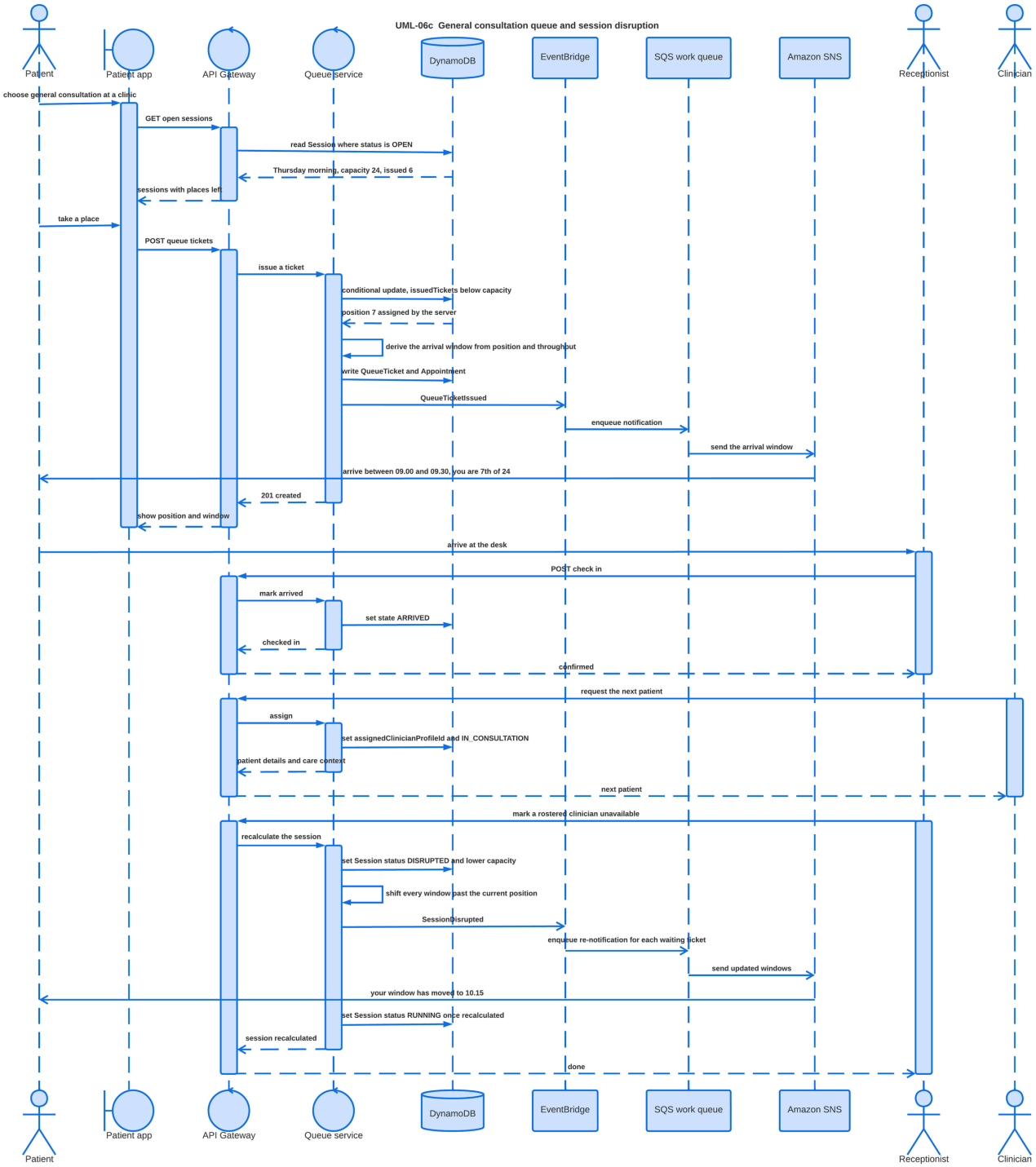


Figure 17. UML-06c general consultation queue and disruption.

- The patient books a ticket in a session. The server assigns the position from the request timestamp and returns the arrival window.
- At the desk, walk-ins join the same queue. Check-in marks arrival; a free clinician takes the next ticket.
- When the session runs late or a clinician is marked unavailable, the queue service recalculates every later window and the notification service texts the patients whose window moved.

7.2 State machines

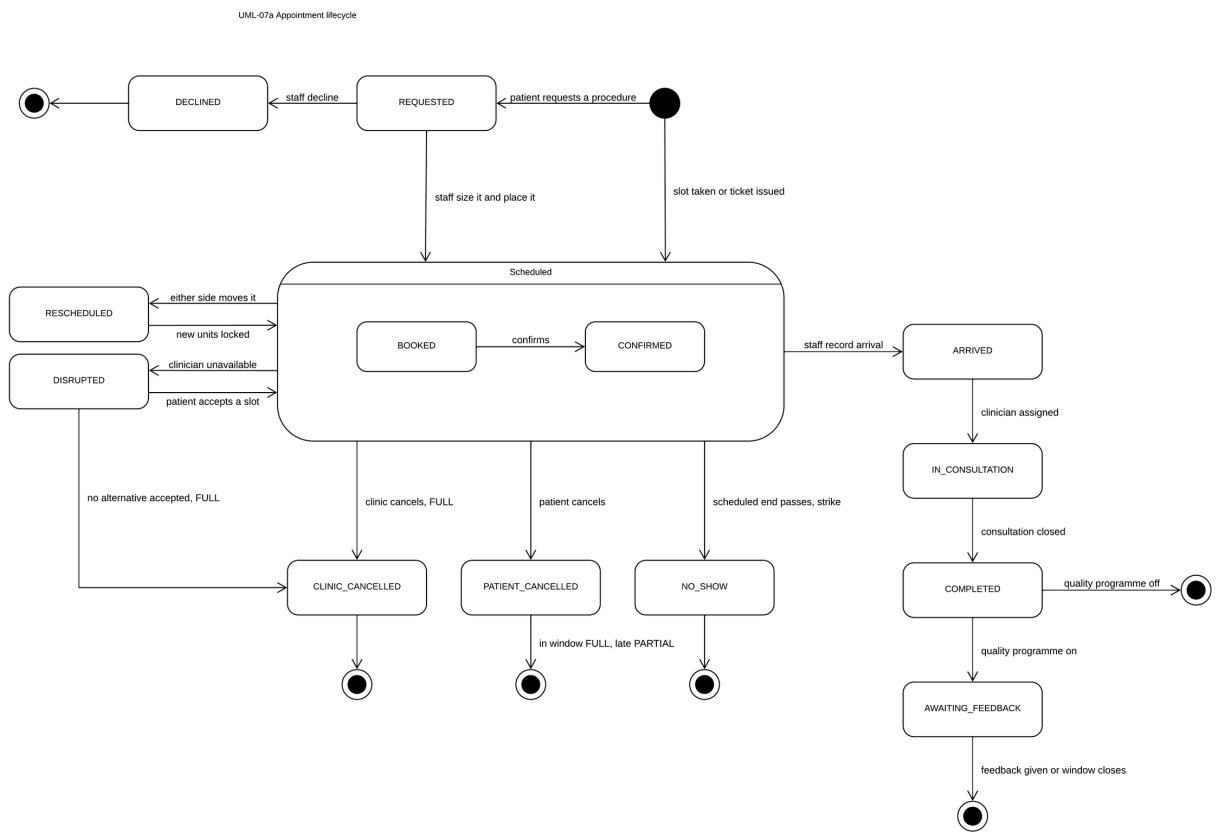


Figure 18. UML-07a appointment lifecycle.

STATE	MEANING	LEDGER OUTCOME AND ENTITLEMENT
REQUESTED	A patient asked for a procedure; no time is held	None yet
DECLINED	Staff declined the request	None
BOOKED, CONFIRMED	Inside Scheduled: a slot or ticket is held; the patient has confirmed where confirmation applies	None yet
RESCHEDULED	Moved by either side; new units locked	None
DISRUPTED	The clinician became unavailable; offers are pending	None yet
ARRIVED	Staff recorded arrival	Attended, NONE
IN_CONSULTATION	A clinician took the patient	Attended, NONE
COMPLETED	Consultation closed	Attended, NONE
AWAITING_FEEDBACK	Completed with the quality programme on; the prompt is open	Attended, NONE
PATIENT_CANCELLED	Cancelled by the patient	In window FULL, late PARTIAL

STATE	MEANING	LEDGER OUTCOME AND ENTITLEMENT
`CLINIC_CANCELLED`	Cancelled by the clinic, or disruption with no alternative accepted	FULL
`NO_SHOW`	The scheduled end passed with no arrival; a strike is counted if strikes are on	NONE

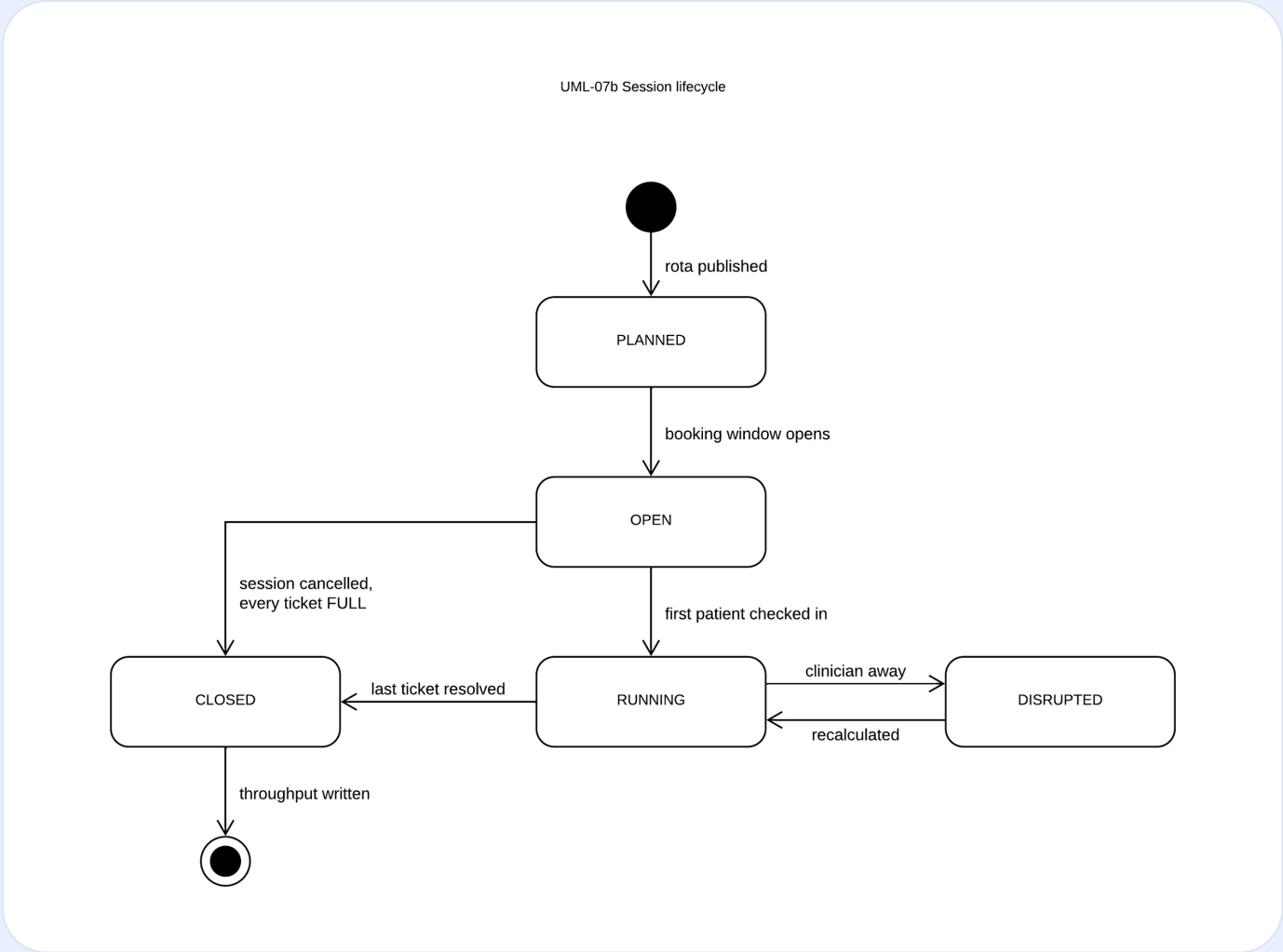


Figure 19. UML-07b session lifecycle: PLANNED, OPEN, RUNNING, DISRUPTED, CLOSED.

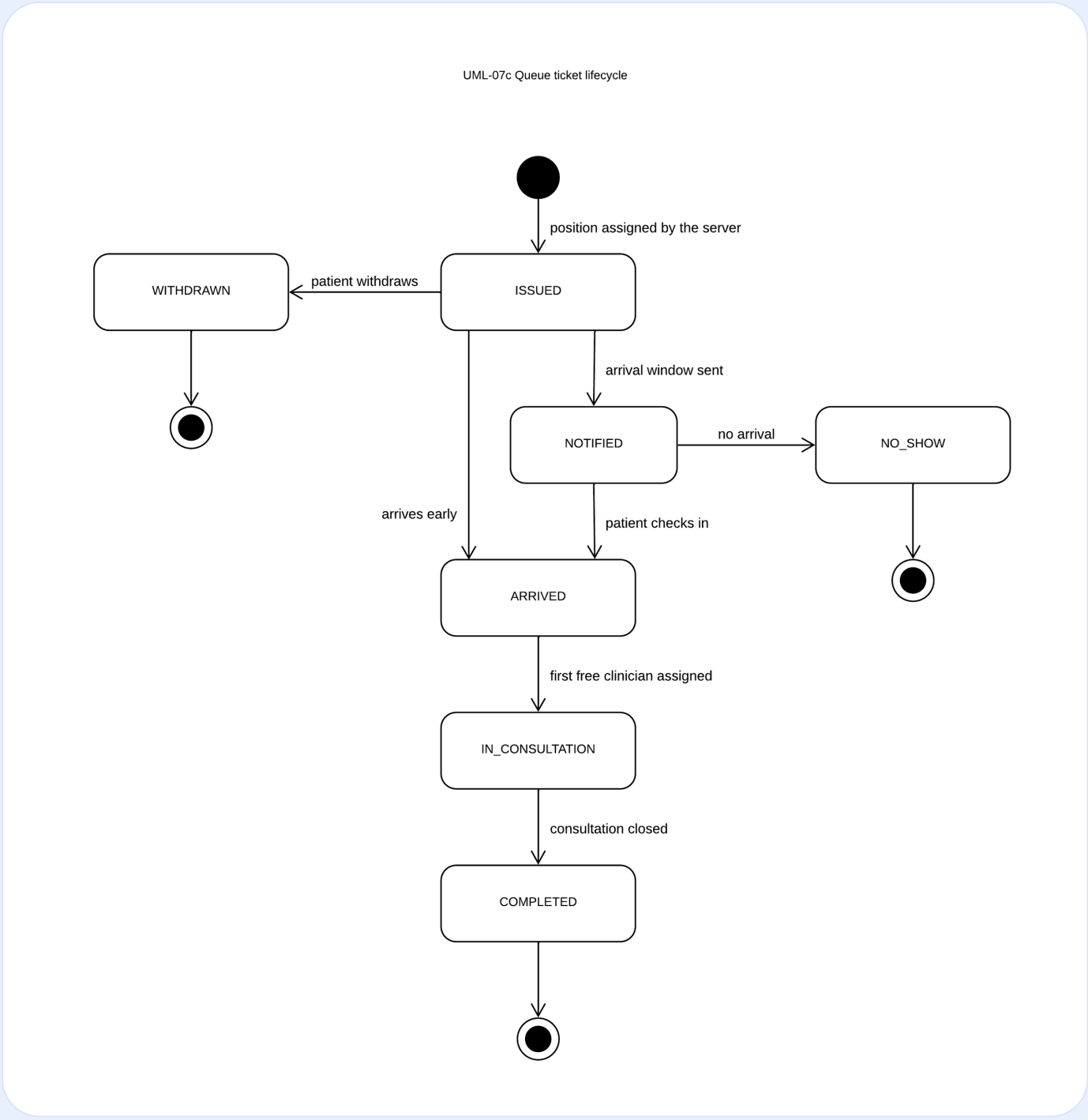


Figure 20. UML-07c queue ticket lifecycle: ISSUED, NOTIFIED, ARRIVED, IN_CONSULTATION, COMPLETED, NO_SHOW, WITHDRAWN.

8 Authorisation

Atria has five roles. Four are staff roles held through a staff membership; the patient role is held through a patient profile. A tenant administrator assigns staff roles when creating the account. A signed-in account receives exactly its roles' permissions: 33 permissions, each granted at one of four scopes, then narrowed by seven guard rules.

SCOPE	MEANING
own	The actor's own records only
clinic	Every record in the actor's clinic
tenant	Every record in the tenant
none	Refused

8.1 Permission matrix

PERMISSION	PATIENT	RECEP.	CLINICIAN	MANAGER	ADMIN
appointment.read	own	clinic	own	clinic	tenant
appointment.create	own	clinic	clinic	clinic	none
appointment.reschedule	own	clinic	own	clinic	none
appointment.cancel	own	clinic	own	clinic	none
appointment.check_in	none	clinic	own	clinic	none
appointment.mark_no_show	none	clinic	own	clinic	none
procedure.size_and_place	none	none	own	clinic	none
procedure.request	own	clinic	clinic	clinic	none
care_context.read	own	none	own	none	none
care_context.write	own	none	own	none	none
availability.read	none	clinic	clinic	clinic	tenant
availability.write	none	none	own	clinic	none
session.publish	none	none	none	clinic	none
session.disrupt	none	clinic	clinic	clinic	none
queue.assign_next	none	clinic	clinic	clinic	none

PERMISSION	PATIENT	RECEP.	CLINICIAN	MANAGER	ADMIN
patient.read	own	clinic	own	clinic	tenant
patient.create	own	clinic	clinic	clinic	none
patient.update	own	clinic	none	clinic	none
staff.create	none	none	none	none	tenant
staff.update_roles	none	none	none	none	tenant
staff.suspend	none	none	none	none	tenant
priority_allowance.grant	none	none	none	none	tenant
clinic.manage	none	none	none	none	tenant
appointment_type.manage	none	none	none	none	tenant
fee_band.manage	none	none	none	none	tenant
region_pack.choose	none	none	none	none	tenant
feature_flag.set	none	none	none	none	tenant
feedback.submit	own	none	none	none	none
feedback.read	own	none	own	clinic	tenant
fee_ledger.read	own	clinic	none	clinic	tenant
audit.read	none	none	none	clinic	tenant
report.read	none	none	own	clinic	tenant
message.send	none	clinic	own	clinic	tenant

8.2 Guard rules

Guard rules run after the matrix grants and can only refuse.

RULE	EFFECT
SELF_SUBJECT_DROPS_STAFF_SCOPE	Where the actor's personId equals the subject's personId, every staff scope is dropped and only the PATIENT row applies. A staff member can never act on their own appointment through staff tooling.
SERVER_ASSIGNED_ORDER	Queue position and slot allocation are derived from the request timestamp on the server. A client supplied position or ordering field is rejected, not ignored.

RULE	EFFECT
NO_SELF_BENEFITING_RESCHEDULE	A staff action that moves another patient later is refused where the actor holds an appointment in the same session or the adjoining slot window. A CLINIC_MANAGER may still perform it, which keeps genuine cases possible.
REASONED_AND_AUDITED	Every staff initiated reschedule or cancellation requires a reason and writes an AUDIT_ENTRY. Where the actor holds an appointment in the affected window the entry carries POTENTIAL_SELF_BENEFIT for the clinic manager's weekly list.
BOUNDED_PRIORITY	Staff priority exists only as a PRIORITY_ALLOWANCE granted by a tenant administrator, with a monthly quota, visible on the admin dashboard. There is no implicit staff priority anywhere.
REGION_REFUSES_FEE_LINK	Setting tenant.ratingAffectsFee is refused where the region pack's allowsRatingFeeLink is false. Enforced in the region pack, not the screen, so it cannot be bypassed through the API. AD-11.
TENANT_ADMIN_NEVER_READS_CARE_CONTEXT	care_context.read is none for TENANT_ADMIN and there is no override. The administrative role and the clinical role are separated on purpose.

8.3 Account provisioning

ROLE	FIELDS REQUIRED AT CREATION
RECEPTIONIST	givenName, familyName, phoneE164, clinicId
CLINICIAN	givenName, familyName, phoneE164, clinicId, specialty, registrationYear, ordreNumber, languages
CLINIC_MANAGER	givenName, familyName, phoneE164, clinicId
TENANT_ADMIN	givenName, familyName, phoneE164, email

- Staff accounts are created only by a tenant administrator. No staff or administrator self-registration exists.
- A staff member signs in to the account the administrator created and lands on that role's home. The session carries only that account's roles; there is no switching between roles at runtime.
- A staff member who is also a patient books through the patient app with a patient sign-in. Staff scope never applies to their own appointments.
- Changing an account's roles takes effect at its next sign-in, and is audited.

9 Interface contract

One REST API over HTTPS with JSON bodies, stage `v1`. Every request carries a Cognito token as `Authorization: Bearer`. Every list is paginated at 50 items or fewer. Booking requests carry an `Idempotency-Key` honoured for 24 hours.

9.1 Identity and accounts

METHOD	PATH	PERMISSION	PURPOSE
POST	/me/profile	patient.update (own)	Complete your profile: confirm names and date of birth, set language
POST	/me/phone/code · /me/phone/verify	patient.update (own)	Send and check the one-time code that sets phoneVerifiedAt
POST	/me/guardian	patient.update (own)	Link a guardian for a patient under 18 and record consent method
GET · PUT	/me/preferences	patient.update (own)	Channels, lead times, proxy recipient, auto-accept for same-day offers
GET · POST	/me/consents · /me/export · /me/erasure	patient.read (own)	Data subject rights
POST	/admin/staff	staff.create	Create a staff account with its roles; fields per role are enforced in the service
PATCH	/admin/staff/{id}/roles	staff.update_roles	Change the roles on an existing account; takes effect at next sign-in
POST	/admin/staff/{id}/suspend	staff.suspend	Suspend an account and revoke its refresh tokens
PUT	/admin/staff/{id}/priority-allowance	priority_allowance.grant	Grant or change a monthly priority quota

9.2 Directory and capacity

METHOD	PATH	PERMISSION	PURPOSE
GET	/clinicians?specialty=&clinicId=	any signed-in	Directory ordered by seniority band. Specialty, qualifications, registration year, languages, clinic; never a score
GET	/clinicians/{id}/slots?date=&typeId=	any signed-in	Free start units for a type's duration and buffer
GET · PUT	/clinicians/{id}/availability	availability.read · availability.write	Weekly template on the grid unit
POST	/clinicians/{id}/unavailability	session.disrupt	Mark a clinician unavailable for a window; starts disruption

METHOD	PATH	PERMISSION	PURPOSE
GET · POST	/sessions	availability.read · session.publish	List or publish sessions with window and rostered GPs; capacity is derived
POST	/sessions/{id}/disrupt	session.disrupt	Shift every ticket's arrival window and notify

9.3 Booking and the day

METHOD	PATH	PERMISSION	PURPOSE
POST	/appointments	appointment.create	Specialist booking: type, clinician, start, channel, intake answers and consent. Locks N units atomically
POST	/sessions/{id}/tickets	appointment.create	General line: issue a queue ticket; position assigned by the server
GET	/tickets/{id}	appointment.read	Live position and arrival window
POST	/procedure-requests	procedure.request	Raise a REQUESTED appointment with no time held
POST	/procedure-requests/{id}/placement	procedure.size_and_place	Size and place a request for the patient to confirm
POST	/appointments/{id}/confirm	appointment.read (own)	Patient confirms a placed procedure or a disruption offer
GET	/patients/me/appointments?range=&from=	appointment.read (own)	Day, week, month or year; five-year default horizon
GET	/appointments/{id}	appointment.read	One appointment within scope
PATCH	/appointments/{id}	appointment.reschedule	Reschedule; staff must give a reason
DELETE	/appointments/{id}	appointment.cancel	Cancel; returns the entitlement band; staff must give a reason
POST	/appointments/{id}/arrival · /no-show	appointment.check_in · appointment.mark_no_show	Record attendance outcomes
POST	/sessions/{id}/next	queue.assign_next	Assign the next ticket to the calling clinician
GET · PUT	/appointments/{id}/care-context	care_context.read · care_context.write	Care context for the patient and the assigned clinician only
POST	/appointments/{id}/referral	appointment.create	Referring facility and clinician, with a presigned upload for the note

9.4 People, quality and money

METHOD	PATH	PERMISSION	PURPOSE
GET · POST	/patients	patient.read · patient.create	Search the register; quick-create a patient at the desk
GET · PATCH	/patients/{id}	patient.read · patient.update	Patient record within scope; strike count when strikes are on
POST	/appointments/{id}/feedback	feedback.submit	Structured answers, free text and a safety or conduct flag; attended visits only
GET	/feedback? clinicId=&clinicianId=&from=	feedback.read	Scoped: own aggregate, clinic aggregate and flagged reviews, or tenant-wide
GET	/fee-ledger?clinicId=&from=	fee_ledger.read	Fee band, XAF amount, outcome and entitlement per appointment

9.5 Institution and oversight

METHOD	PATH	PERMISSION	PURPOSE
GET · POST · PATCH	/admin/clinics · /admin/appointment-types · /admin/intake-forms	clinic.manage · appointment_type.manage	Institution configuration
GET · POST · PATCH	/admin/fee-bands	fee_band.manage	Per-practice baselines from the ministerial schedule
PUT	/admin/region-pack	region_pack.choose	Choose the region pack; refused once the first appointment exists
PUT	/admin/feature-flags	feature_flag.set	Quality programme, strikes; ratingAffectsFee refused where the pack forbids it
GET	/audit?actor=&action=&from=&to=	audit.read	Read-only audit trail
GET	/reports/attendance?by=channel · /reports/reschedules/weekly	report.read	Attendance and no-show by channel; weekly reschedules grouped by actor
GET · POST	/messages?status=	message.send	Message log with failures first; resend

9.6 Errors

STATUS	WHEN
400	The body fails schema validation

STATUS	WHEN
401	No token or an expired token
403	The matrix or a guard rule refuses; the body names the rule
404	Not found within the caller's scope
409	A slot unit is taken, or the state no longer allows the action
422	A business rule refuses: region locked, fee link refused, reason missing, notice period passed
429	Rate limit exceeded
500	Unexpected failure, with a correlation id

Errors return `{"error":{"code","message","rule","correlationId"}}` .

9.7 Messaging interfaces

INTERFACE	CONTRACT
SMS	SNS transactional SMS to E.164 numbers with the registered sender ID. Templates in French and English fit one GSM-7 segment where possible. A provider adapter allows a third-party SMS or voice provider whose key is held in Secrets Manager.
Email	SES from a verified domain with DKIM, SPF and DMARC, through a configuration set that publishes delivery, bounce and complaint events.
Identity	Cognito with Google on the patient client only, authorisation code flow, scopes openid email profile, and optional birthday and phone permissions that may be declined.

10 Delivery and operations

10.1 Repository and branches

PACKAGE	CONTENTS
spec	The specification as data: roles and the permission matrix, the entity model, the key design and the lifecycles. Imported by everything that runs on AWS and generated into the web client, so there is one definition of each rule
infra	CDK v2 Python app with the stacks in section 5.3 and per-environment configuration
backend	Lambda handlers in Python, domain logic, the authoriser and the permission matrix as code
web	React web app for every role
app	Flutter app for patient and staff
docs	Architecture decision records, the API contract and the constraints
tools	Operational scripts, including the end-to-end check against a deployed environment

Each feature is built on its own branch and merged by reviewed pull request into the integration branch. Releases are tags on the main branch.

10.2 Pipeline

STAGE	RUNS	GATE
Pull request	Lint and format checks, strict type checking, unit tests with a coverage floor, the assertions on every synthesised template, CDK synth, the check that the web client's generated copy of the specification is current, Flutter analyze and test, web build and test, secret scan	Any failure blocks the merge
Integration	Deploy all stacks to development, integration tests on every endpoint	Failure blocks promotion
Release	Deploy to staging, end-to-end booking, queue and cancellation tests on both clients	Manual approval
Production	Deploy the tagged release	Manual approval; authorisations held

10.3 Testing

- Domain logic has at least 80 per cent line coverage, including every guard rule and every entitlement outcome.
- Concurrency tests book the same unit from parallel clients and assert exactly one success.
- Authorisation tests call every endpoint with every role and assert the matrix and guard rules.
- Reminder tests assert that a cancelled or moved appointment sends nothing and that a due reminder publishes within two minutes of its time.

- Each test names the requirement it verifies, so the verification column of the requirements specification can be read back from the suite rather than asserted.
- Infrastructure is tested by assertions on the synthesised template, stating properties such as every route carrying an authoriser and care context holding its own key, rather than by a snapshot that changes on every unrelated edit.

10.4 Alarms and runbooks

ALARM	THRESHOLD	FIRST RESPONSE
Dead letter depth	Any DLQ above 0 for 5 minutes	Inspect the parked message, fix, redrive from the console runbook
Reminder failure rate	Above 5 per cent of sends in 1 hour	Check carrier status and sender ID; phone affected patients from the message log
API 5xx rate	Above 1 per cent for 5 minutes	Check the latest deployment and X-Ray traces on the booking path
Lambda throttles	Any throttle for 5 minutes	Raise reserved concurrency or smooth the burst through SQS
DynamoDB throttles	Any throttled request	Check hot keys on slot locks and the outbox shard
SES bounce rate	Above 3 per cent daily	Pause bulk email, clean addresses
SMS daily spend	Tenant cap reached	Non-essential SMS paused; reminders continue within cap
Budget	50, 80, 100 per cent of monthly budget; anomaly detected	Review Cost Explorer and the scheduler for misfires

10.5 Backup and recovery

- Point-in-time recovery on every table gives a recovery point within five minutes; a full restore completes within four hours.
- S3 referral documents are versioned with lifecycle expiry matching the appointment retention.
- Dead letter queues retain messages for 14 days; redrive is a documented console action.

11 Open technical issues

Known technical issues that are not yet resolved, each with the point by which it must close.

ID	ISSUE	IMPACT	RESOLUTION	BY
OI-02	Production region and the authorisation to transfer personal data abroad.	Development stays on synthetic data until decided.	Choose the production region before real patient data; apply for transfer authorisation.	Before Phase 3
OI-04	SMS sender ID registration for MTN, Orange and Camtel differs by region.	Reminders may be filtered or unbranded.	Register sender IDs for the production region; test on all three networks.	Before Phase 3
OI-11	Prior authorisation to process minors' sensitive data.	A launch gate for the guardian flow.	Apply with the processing authorisation; the flow is built and demonstrable.	Before Phase 3
OI-12	No statutory retention period for Cameroonian medical records was found.	Ten years retained, five shown is a proposal.	Confirm with counsel; values are tenant settings.	Before Phase 3
OI-13	Voice fallback provider for Cameroon is not selected.	Voice reminders unavailable.	Evaluate providers behind the messaging adapter.	Phase 3